Universität Münster

University of Münster
School of Business & Economics

Abishek Sivakumar, Aizhan Kanybekova, Chalisa Ivanka Salsabila, Lorenzo Lazarus

ERP-inspired Nursing Home Planning System

Seminar Thesis
in the context of the seminar "Integrated Research Seminar"

at the Chair for Information Systems and Information Management
(Universität Münster, Münster)

Supervisors: Prof. Michael Räckers, Prof. Tobias Brandt

Tutor: David Nowak

Presented by: Abishek Sivakumar
Aizhan Kanybekova
Chalisa Ivanka Salsabila
Lorenzo Lazarus

Date of Submission: 2025-08-02

Content

Figures 3
Abbreviations 4
1. Introduction 1
2. Theoretical Background 3
2.1. Health Management in Europe 3
2.2. Healthcare planning in Germany 3
2.3. ERP, Business Intelligence, and TOGAF Overview 5
2.4. Sense-making of Website 8
3. Methods 10
3.1. Activity 1 - Identify a problem and motivate 10
3.2. Activity 2 - Define objectives of the solution 12
3.3. Activity 3 - Design and development 12
3.3.1. TOGAF ADM and the system design alignment 13
3.3.2. Rights and Roles (RBAC) 20
3.3.3. User Interface 22
3.4. Activity 4 - Demonstration 27
3.5. Activity 5 - Evaluation 27
3.6. Activity 6 - Communication 28
4. Results 29
5. Discussion 35
5.1. Theoretical and Practical Implication 35
5.2. Legal Implication 36
5.3. Design Implication: Alternatives Analysis on Standard Software or Custom Development 36
6. Conclusion 39
References 41
Appendix 54
Appendix 1: Interview Transcript 54
Appendix 2: Questions Guidelines 54

Figures

Figure 3.1 UML process flow diagram of the current nursing home planning process **11**
Figure 3.2 Actors and processes identification **13**
Figure 3.3 UML use case diagram **14**
Figure 3.4 UML Component Diagram (Version 1) **16**
Figure 3.5 RBAC matrix **21**
Figure 3.6 Log In page **22**
Figure 3.7 Nursing home dashboard **23**
Figure 3.8(a) Step 1 of a new application nursing home creation **23**
Figure 3.8(b) and **(c)** Step 2 and 3 of a new application nursing home creation **24**
Figure 3.9 Screening dashboard for government **25**
Figure 3.10 Report creation dashboard of the government **25**
Figure 3.11 Report dashboard for the health ministry **26**
Figure 3.12 Document overview **26**
Figure 3.13 Informants' table **27**
Figure 4.1 UML Component Diagram (Version 2) **30**
Figure 4.2 The application form with tooltip **31**
Figure 4.3 Layered Architecture Model **33**

Abbreviations

AD	Active Directory
ADM	Architecture Development Method
BI	Business Intelligence
BBD	Barone Budge & Dominick
DMS	Document Management System
DRK	Deutsches Rotes Kreuz
DSD	Dynamic Separation of Duties
DSR	Design Science Research
DSRM	Design Science Research Methodology
EA	Enterprise Architecture
ERP	Enterprise Resource Planning
ETL	Extract Transform Load
EU	European Union
GIS	Geographic Information System
GDPR	General Data Protection Regulation
IAM	Identity and Access Management
IT	Information Technology
LTC	Long-Term Care
MD	Medical Service
NIST	The National Institute of Standards and Technology
NPM	New Public Management
PPP	Public-Private Partnership
RBAC	Role-Based Access Control
SGB	Sozialgesetzbuch - Social Code
TOGAF	The Open Group Architecture Framework
UI	User Interface
UML	Unified Modelling Language
UX	User experience

1. Introduction

Germany's healthcare system faces significant challenges as demographic change accelerates and economic pressure rises (*Demography Report*, n.d., p. 9). Among European Union members Germany has the highest healthcare expenditure reaching 12,6% of its GDP in 2022 (Schmitt et al., 2022, p. 14; *Health Expenditures in the EU 2022 - German Federal Statistical Office*, n.d.). The demographic imperative is intense: Germany's population aged 65 and older is expected to grow to 20.9 million by 2040, representing one fourth of a population (*Population by Age Group and Variant until 2070*, n.d.). Thus the number of people requiring long-term care (LTC) from five million people in 2021 is projected to grow to approximately 5.6 million by 2035 (*Long-Term Care Projection*, n.d.).

Due to the increasing demand, Germany has undertaken policy reforms to reduce long-term care gaps (Jensen et al., 2024, p. 1). One reform, *Pflegestärkungsgesetz III*, streamlines the collaborative practice in Germany's healthcare. This regulation enables municipalities to establish local LTCs that cover local needs and allows adequate resource allocation to set the maximum number of patients assigned to nursing staff and ensures tailored service. However, despite its substantial investments, Germany's healthcare outcomes stay disproportionate to its costs (OECD, 2019, p. 22). The expected benefits of collaborative governance are hindered by complex administration and numerous actors that are delegated by decision-making power, including self-governing bodies failing to deliver real-time data pertinent to healthcare service (OECD, 2019, pp. 9-15; Schmitt et al., 2022, p. 26).

These challenges are most noticeable in nursing home planning, where they stem from the absence of a unified system for stakeholder interaction. The nursing home planning process involves stakeholders such as nursing homes, insurance associations, government agencies and the Health Ministry. The current reliance on paper or email-based communication, combined with outdated IT infrastructure, leads to poor coordination and passive stakeholder participation.

Despite these challenges and the disintegration of governance in the healthcare system, governance remains vital, following the fundamentals of New Public Management (NPM) and Whole-of-Government views that promote intersectoral collaboration (Hurrelmann, 2019, pp. 23-26; Andersson, 2014, p. 41). From the New Public Management (NPM) perspective, smart governance in healthcare fosters

healthcare as part of Whole-of-Government (WoG), meaning that healthcare system decision-making must include various actors participatory governance, and adaptive and resilient governance (Kickbusch & Gleicher, 2014, p. 89-160, WHO, 2014). A study probing the Neustart project using the TAPIC (Transparency, Accountability, Participation, Integrity and Capacity) framework argues intersectoral collaboration and self-governing bodies should not be eliminated and should be an integral part that supports transparency in the healthcare system (Schmitt et al., 2022, p. 19). Therefore, following the current healthcare transformation (§ 1 SGB XI) and NPM perspective, this research proposes a unified digital platform to streamline nursing home planning (see Section 3.1). Using a Design Science Research (DSR) methodology, this study develops a conceptual model of an underlying architecture of the proposed platform based on an Enterprise Resource Planning system to address current inefficiencies (see Section 3.2). Additionally, this paper contributes to the underexplored field of ERP integration in the public sector practices.

2. Theoretical Background

2.1. Health Management in Europe

The shift in public management has catalyzed new service delivery in a digital manner, aiming for more efficient, transparent, and customized approaches. It advocates the amalgamation of public and private actors to improve public administration (Hood, 1991, p.5)**.** The Omaolo platform is the epitome of healthcare governance exerting public-private partnership (PPP) by unifying the public sector software within the ecosystem that ensures scalability, open communication, and transparency (Kolehmainen et al., 2024, p. vi-viii; Stoker, 1998, p. 17). Hurrelmann (2019) elaborated further that the governance model and practices must align with local and global prevailing needs and upcoming demands (pp. 23-26). However, this creates escalating complexity caused by regulatory compliance tightly surrounding public sector practices and other multiple factors that need to be included in digitalizing service. As a result, TOGAF becomes handy in aligning the proposed design with policy and governance in Germany, especially when the context where the system is implemented is tightly regulated and complex.

2.2. Healthcare planning in Germany

In 1995 in response to growing need for elderly care, the German government established long-term care insurance or "Pflegeversicherung", which plays a crucial role in funding care services (*Pflegeversicherung*, n.d.). The healthcare planning in Germany is decentralized, involving collaboration between federal and state governments, healthcare providers, and insurance associations. The long-term care planning, or in this case nursing home planning is guided by regulations such as Social Code 5 (SGB) and SGB 11 on federal level, and Home Law/Act (HeimG) on state level.

SGB 5 governs Germany's statutory health insurance (GKW) system and outlines the responsibilities of health insurance funds, public authorities, and service providers in organizing and financing healthcare services (*SGB V Statutory Health Insurance - SGB V Social Code*, n.d.). According to §69-71 insurance associations participate in negotiation and approval of nursing home service mandates by collaborating with government agencies to ensure care quality and efficiency. §275 mandates the involvement of the Medical Service (MD) in reviewing the necessity and

appropriateness of services. Additionally, §90 establishes regional cooperation structures where health insurers, public health authorities, and care providers assess care needs and develop service mandates. These clauses reinforce the inclusion of the insurers and evaluators as core stakeholders in the platform, however with well defined access rights and responsibilities.

SGB 11, on the other hand, provides the legal framework for Germany's long-term care insurance and is directly associated with operation, authorization and the evaluation of the nursing homes. According to §72 care insurance funds are only allowed to offer care through care facilities (nursing homes) that have a valid Versorgungsvertrag or supply contract, with them (care facilities establish contracts with care insurance funds to be eligible for reimbursement). This way it ensures that only vetted and contractually bound providers can deliver care, sets clear expectations for service quality and labor standards, and provides a legal foundation for the protection of both care recipients and employees. Furthermore, §75 of SGB 11 ensures uniform conditions for care provision at the state level by Rahmenverträge. Rahmenverträge or framework agreements define the scope and content of care services, establish general conditions for care contracts (such as requirements for contract fulfillment, cost coverage, invoicing documentation, reporting), and set standards for the staffing and equipment of care facilities. To ensure quality of services §112, §113 and §114 of SGB 11 create a comprehensive quality assurance framework in long-term care. §112 establishes that care providers' are responsible for quality of their services and are obliged to participate in quality assurance processes under §114 of SGB 11, while §113 (Standards and Principles for Securing and Developing Care Quality) regulates the specific agreements (Qualitätsvereinbarungen) between care insurance funds and nursing homes about quality management and assurance. These agreements should specify requirements for care quality and documentation, necessary qualifications and staffing, procedures for quality assurance and management, and continuing education of staff. Additionally, Qualitätsvereinbarungen defines the standards for quality inspections and the involvement of of external bodies to ensure systematic monitoring and improvement of care quality Those quality inspections and external bodies are specified under §114, where external quality inspections are conducted by MD or similar bodies to assess whether the care provided meets quality standards and are based

on the requirements set out in §112 and §113 (*SGB XI Soziale Pflegeversicherung - SGB XI Sozialgesetzbuch*, n.d.).

Overall, SGB 11 provides the structure for care, quality and funding of nursing homes while SGB 5 ensures that residents continue to receive necessary quality medical care within these facilities.

In addition to federal requirements, state laws or Landesheimgesetze regulates the day-to-day operations of care facilities, including operational standards, licensing, and supervision of nursing homes. These state laws differ from state to state and may have different names; for instance, in North Rhine-Westphalia there are 5 HeimGesetz. These state laws ensure that nursing homes and other care facilities meet minimum requirements for safety, staffing, and quality; however, unlike the SGB 11 do not govern the financial entitlements or insurance aspects of care (*Home Laws of the Federal States | BIVA Nurgeschutzbund*, 2014; *HeimG—Home Law*, n.d.; Suhr et al., 2018).

2.3. ERP, Business Intelligence, and TOGAF Overview

In recent years, due to the advancement of Information Technologies, Enterprise Resource Planning (ERP) systems are being widely used in private sector organizations, primarily for the streamlining of business processes. Cai & Zhu (2015, p. 1) argue that ERP allows organizations to analyze large volumes of data to improve service quality. Additionally, ERP systems allow for the integration and automation of internal processes across departments, creating centralized and consistent information flows (Shang & Seddon, 2002, p. 2).

Enterprise Resource Planning (ERP) systems and Business Intelligence (BI) tools are increasingly being integrated to enhance organizational efficiency and informed decision-making. ERP systems are designed to collect, store, and manage structured data across various functional domains such as finance, supply chain, and human resources. However, they often lack the analytical capabilities required to derive deeper insights from this data. Business Intelligence complements ERP by enabling organizations to convert raw operational data into actionable insights through dashboards, reports, and visual analytics (Chou et al., 2005, p. 1). According to Nofal and Yusof (2013, p. 3), integrating ERP and BI empowers organizations to streamline operations while becoming more agile and competitive. BI tools offer advanced data interpretation capabilities that ERP systems alone cannot provide. Moussas et al. (2024,

p. 11) further highlight that BI supports both strategic and operational decision-making by providing real-time visibility into organizational performance. Hou and Papamichail (2010, p. 1) affirm that the combined use of ERP and BI leads to significantly improved decision outcomes compared to using ERP in isolation.

Recent studies found that ERP offers real-time visibility of operational routine while streamlining and optimizing important tasks by integrating various business components (Shbail et al., 2024, pp. 121-124). This supports the demand for data-driven decision-making (Javaid et al., 2022, p. 204). As a consequence, data becomes the integral part that requires maintenance for keeping its quality and integrity intact (Cavallo et al., 2021, p. 265; Zheng and Khalid, 2022, p. 4; Abd Ali and Smaoui, 2023).

While ERP systems have proven effective in many private sector contexts, ERP's implementation in the public sector has produced mixed results. On one hand, research shows that in public sector institutions, ERP systems contribute to more structured data management by ensuring reliable and easily accessible information flows across different administrative units (John, 2024, p. 10). On the other hand, public-sector deployments face challenges due to bureaucratic complexity, rigid governance structures, and diverse stakeholder requirements. ERP systems have been introduced in various public sector domains such as education, healthcare and defence across municipal, regional, state and national levels to modernize public administration (Kelemen, 2014, p. 6). Such reforms have the tendency to reduce fragmentation, enhance accountability, and align internal operations with governance objectives and evolving public needs. Although recognizing the difference between sectors, the majority of public agencies have borrowed ERP implementation methods first designed for the private sector and tried to adapt them for the unique operating and legal needs of the public sector (Wagner & Antonucci, 2009, p. 4). Such adaptations typically include careful planning, establishment of roles, and synchronization with long-term strategic goals—areas where enterprise architecture frameworks can play a crucial role.

To ensure strategic alignment and modular system design, several studies recommend the use of enterprise architecture frameworks. Hindarto & Wedha (2023, p. 1) claim that aligning ERP systems with Enterprise Architecture (EA) is essential for maximizing organizational efficiency, as EA provides a structured framework that enhances ERP integration, scalability, and adaptability in dynamic business environments. Public sector procurement units often face data duplication and system

fragmentation due to siloed ICT architectures, which Finland addresses through a mandatory EA framework to promote interoperability (Ghezzi et al., 2024, p. 1). Recent studies have shifted attention toward understanding not just the potential of EA in the public sector, but also the specific mechanisms through which it creates value. For instance, research based on Swedish government organizations has developed a conceptual framework identifying key EA activities, organizing principles, and applications that contribute to public value creation, offering strategic guidance for public managers (Sundberg et al., 2023, p.1). Developed by The Open Group, The Open Group Architecture Framework (TOGAF) provides a structured methodology for designing, implementing, and governing complex systems. Its Architecture Development Method (ADM) is particularly relevant in public sector contexts, offering a phased approach to system design that supports customization based on organizational processes, stakeholder needs, and data structures (Triyanto & Supriyanto, 2024, p. 1). Using TOGAF ADM and national architectural guidelines, an Enterprise Architecture blueprint was developed for the electricity sector in West Java, producing structured outputs across key architectural domains including business processes, data, applications, and services (Kencana et al., 2022, p. 11).

TOGAF ADM has various phases. The Preliminary Phase sets the foundation by defining the scope, architectural principles, and governance structure for the architecture effort. It ensures that all stakeholders are aligned on the methodology and objectives (The Open Group, 2022). The Requirements Management phase in TOGAF ADM acts as a continuous thread that runs through every stage of the architecture process. Its role is to make sure that all system requirements are clearly defined, tracked, and updated as things evolve. It works alongside each ADM phase to ensure the architecture stays aligned with organizational goals and can adapt smoothly to changes in needs or context. Phase A is the Architecture Vision that focuses on the expected values of the system. It identifies key stakeholders, defines the scope of the solution, and establishes a shared understanding of what the architecture seeks to achieve. Phase B is the Business Architecture that models the organization's business processes and stakeholder interactions. It details the operational aspects that the architecture must support and is typically expressed through process models and use case diagrams. Phase C is the Application Architecture that defines the software components, their responsibilities, and interactions. This phase translates business needs into application logic and

specifies how different modules of the system will function together. Phase D is the Technology Architecture that delves into the underlying infrastructure that supports the application layer. This includes databases, servers, network configurations, and cloud services. It ensures that the system will be scalable, secure, and technically feasible. Phase E, which is the Opportunities and Solutions phase, focuses on identifying the essential tools and components required for implementation of the system while outlining integration strategies and evaluating the use of third-party or existing technologies. This phase also serves as a critical link between the architectural design and its practical execution. Phase F is the Migration Planning which develops a roadmap for transitioning from the current state to the future system. This includes identifying dependencies, managing risks, and planning for data migration or process changeovers. Phase G, Implementation Governance, establishes mechanisms to oversee the execution of the architecture. This includes compliance monitoring, project management alignment, and stakeholder reviews to ensure proper delivery. Phase H is Architecture Change Management that prepares the organization for future adaptations. It outlines how the system can be updated in response to evolving policies, technologies, or stakeholder needs.

2.4. Sense-making of Website

Since Germany tends to digitalize traditionally or replicate the existing system, this research approaches the use case by digitalizing the current system using a website. Due to different interests that need to be catered to, digital health governance must ensure inclusivity, potentially promoting sustainable business models in digital healthcare planning (Smith et al., 2024, p. 15). Different usage is likely to become a barrier in public e-services communication that can affect sense-making and trust, leading to failure in government sites. This is where the concept of sense-making of the e-government becomes critical to increase the user-centric design.

Sensemaking gives rationality to the design demonstrated through images to guide and help users interpret unequivocality and meanings (Weick, Sutcliffe, & Obstfeld, 2005, p. 409). Recent study elaborates that sensemaking is leveraged by digital nudging, guiding users with simple elements, clear signs, FAQs, and feedback mechanisms to reduce cognitive load (Gonin and Neumann, 2025 p. 679) . In Germany's public administration, explainability in digital services has been embedded

to increase ease of use, frictionless navigation, and efficient operation (Fahr et al., 2025, pp. 553-555; Gräfe et al., 2024, pp. 8-9).

Since the design is limited on practical tests, this research embraces the intuitive design approach by Krug's proposal, as it is the most practical compared to other affordance and usefulness theories. Krug (2014) emphasizes self-explanatory buttons or signs to fulfill 'satisficing' design that represents average users (p.54). Satisficing, referring to Simon (1957, 1959) and Williamson (1958), is built as a response to bounded rationality, in which users tend to choose the option that meets the minimum criteria or is 'good enough.' Hence, satisfaction is not only helpful in the sense of providing a reasonable option but it also accommodates people with limited cognitive skills. With Krug's proposal, the design aims for clear navigation with clear menus, a consistent layout, and breadcrumbs, usually shown as the 'back' button. Other than the backtracking navigation pane, Krug (2014) emphasizes questions for a clearer navigation pane for orientation and exploration, which are "where can I?" and "Where can I go?" respectively (pp. 63-88). This concept provides clarity instead of unnecessary cognitive load pertaining to the common understanding of sensemaking by preceding authors.

3. Methods

The Design Science Research Methodology (DSRM) is known for its ability to connect research and professional practices by proposing artifacts that solve a specific problem (Hevner et al., n.d., p. 77; Lapão et al., 2017, p. 3; Peffers et al., 2007, p. 47). In order to create the artifact - in this case, a platform for nursing home planning - this study follows DSRM proposed by Peffers and colleagues (2007). Peffers and colleagues' methodology structures the research process into six activities: problem identification and motivation, objectives of a solution, design and development, demonstration, evaluation, and communication.

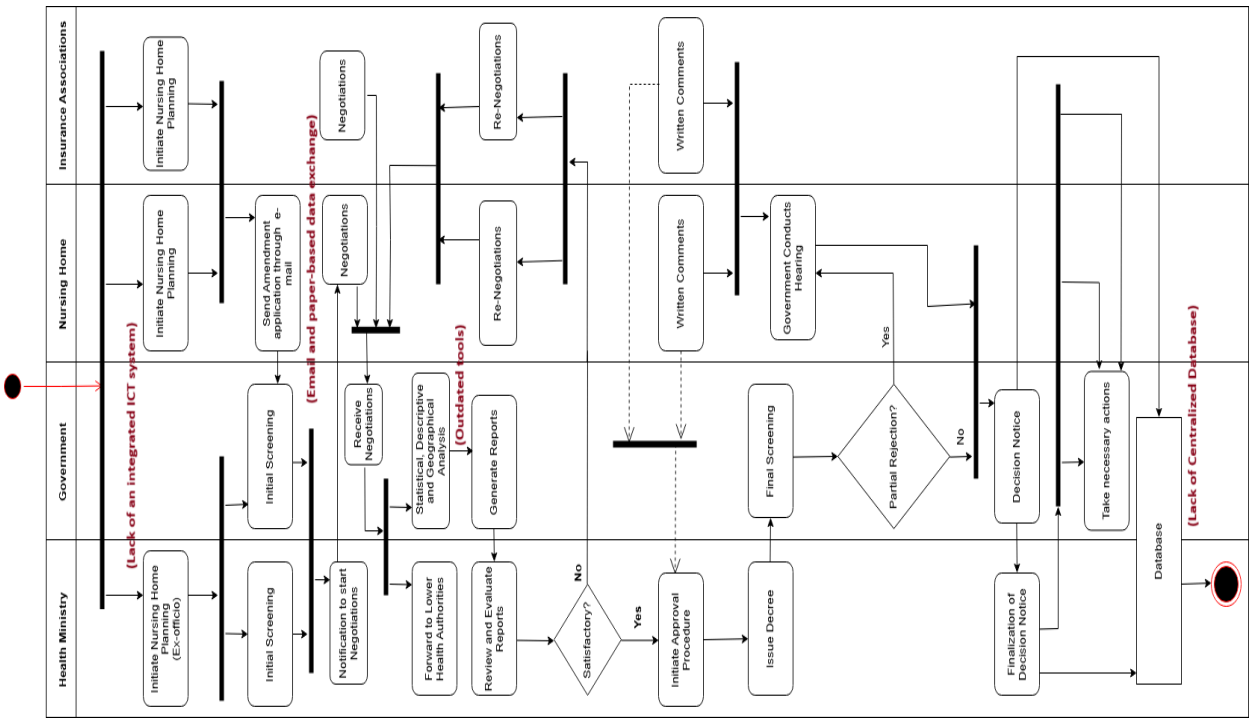
3.1. Activity 1 - Identify a problem and motivate

The foundational problem and its contextual importance were already described in detail in the introduction (see Section 1). This sub-chapter describes the process and identifies weaknesses in detail. The planning of the nursing homes is a legally mandated and multi-stakeholder process, involving the federal state health ministry, government agencies of the state, insurance associations, and nursing homes. The process flow diagram (Figure 3.1) shows the full process from the start till the end, and marks the problems identified.

The process begins when nursing home planning is initiated either by government authorities or through application submission by nursing homes or insurance associations via email. Afterwards the screening by the government agency takes place to ensure the application complies with the requirements. After screening, a one-month period begins for negotiations between nursing homes and insurance associations. These negotiations focus on quality criteria, needs, efficiency of the nursing home (see Section "*Healthcare planning*"), which can last up to six months. The health ministry and the government agency of the relevant state can be invited to these discussions but must remain neutral. Once the negotiations are finalized, its results are sent to the corresponding government authority, which can forward them to other relevant stakeholders. The government then conducts descriptive, statistical, and geographical analyses, and creates a comprehensive report. This report is then sent to the state health ministry to review and evaluate it . If the health ministry is dissatisfied, new negotiations are instructed. However, if satisfied, an approval procedure begins,

where involved stakeholders can submit written comments on the report. Finally, the health ministry makes a decision, instructing the responsible government authority to conclude the proceedings. The responsible government agency then makes final screening and evaluation, issues a decision notice in case of approval, or conducts a hearing in case of partial rejection. Upon the finalization the decision notice is created, which is then available to negotiators and sent to the health ministry.

Figure 3.1 UML process flow diagram of the current nursing home planning process



The study found several weaknesses in the planning process, stemming primarily from a lack of a unified digital system and heavy reliance on manual procedures. First of all, the process is initiated manually, where stakeholders submit applications from different disconnected systems. In addition to this, the data is not centrally stored in one system, which limits the data management and raises concerns about the management and traceability of the individual applications. Furthermore, this lack of central repository is a bottleneck for effective communication between stakeholders, some of whom are currently only passively involved in the process.

Secondly, the tools used in the evaluation and analyses are outdated and partially unsupported, particularly GeoTools used for regional distribution analysis. The quality of evaluations and analyses' reports is highly dependent on the effectiveness of the tools used for these operations. Moreover, the nursing home information system, which also

does not cover all desired functionalities, will be phased out at the end of the year. These pinpointed challenges in the current process will guide the objectives of the solution proposed by this research.

3.2. Activity 2 - Define objectives of the solution

Objectives of the solution are derived directly from the weaknesses. Based on these shortcomings and their optimization potential, the new nursing home planning system should be designed to be seamless and more efficient through a unified system - web-platform in this particular case. Specifically, the artifact should:

- digitize and centralize application and negotiation workflows,
- improve transparency through real-time application tracking,
- support data-driven decision-making with integration of BI tools for conducting statistical, descriptive and geospatial analysis,
- ensure structured governance by implementing Role-Based Access Control (RBAC) to clarify each stakeholder's responsibilities and system permissions.,
- develop a technical architecture that support the integration of legacy systems and support future scalability,
- and finally, be procedurally compliant.

3.3. Activity 3 - Design and development

The artifact is a conceptual platform inspired by ERP system design principles. Therefore, the design and development of system architecture was guided by the Open Group Architecture Framework (TOGAF) Architecture Development Method (ADM) methodology. TOGAF ADM consists of iterative phases including preliminary phase, architecture vision, business architecture, information systems architecture, technology architecture, opportunities and solutions, migration planning, implementation governance, and architecture change management.

In this project, the primary focus is on the *application*, *data*, and *technology layers*, aligning with the system requirements and stakeholder interactions in the context of public healthcare governance.

3.3.1. TOGAF ADM and the system design alignment

The Preliminary phase involved choosing TOGAF as the enterprise architecture framework to structure the digital transformation of nursing home planning. It was tailored to the public sector healthcare context by setting key design principles: modularity, data interoperability, and role-based access. The architecture is intended to address the fragmentation of communication and inefficiencies in document and application handling observed in the current process. The overall architecture follows TOGAF ADM helping to ensure a modular, scalable, and interoperable system (Ji et al., 2022, p. 2).

Requirements Management Phase defined the system's core objectives, such as the inter-organizational coordination, transparent application handling, evaluation automation, data centralization, and integration with legacy infrastructures. These formed the baseline for scoping and visioning activities in Phase A.

Phase A – Architecture Vision: The proposed nursing home system addresses existing pain points in the planning of nursing homes, such as disconnected workflows, lack of centralized oversight, and document duplication. The vision is to build an interoperable, user-centered digital system that allows stakeholders from local care providers to federal health ministries to coordinate efficiently. The ERP acts as the operational backbone, while BI analytics support demographic forecasting and informed policymaking.

Phase B – Business Architecture: The business architecture was derived from stakeholder analysis and process understanding and captured via a UML use case diagram (Figure 3.3). The roles of each stakeholder in the new system are as follows (Figure 3.2):

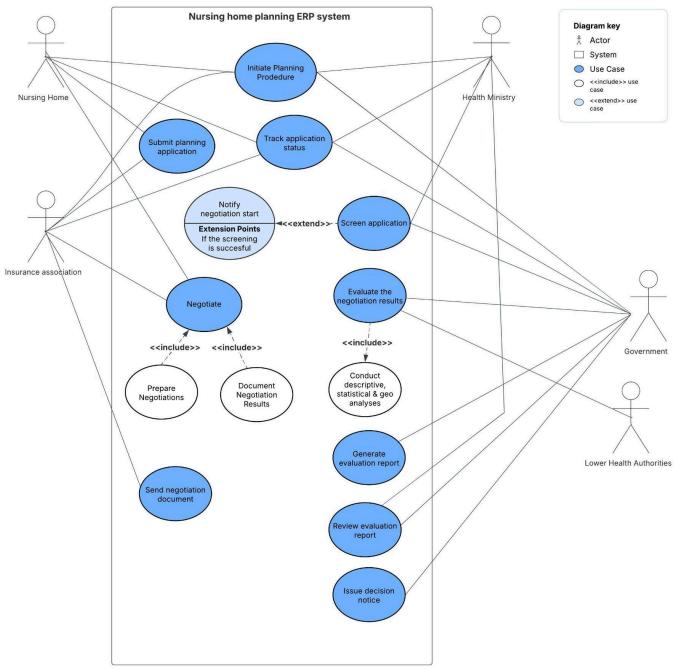
Figure 3.2 Actors and processes identification

Process category	**Process Identified**	**Actor**
Application	Submit planning application	Nursing home Insurance association
	Track application status	Health ministry Nursing home Insurance association

	Screening application	Health Ministry Government agency
Negotiation	Prepare negotiation	Insurance association
	Document negotiation results	Insurance association
	Send negotiation document	Insurance association
	Evaluate negotiation results	Government Lower health authorities
Reporting	Generate evaluation report	Government
	Review evaluation report	Health Ministry
Decision	Issue decision notice	Government

The use case diagram (Figure 3.3) represents how the stakeholders interact with and within the new system. Firstly, the emphasis of the study is done on the process flow that begins with an initiation by either a nursing home or insurance association. Therefore, the subsequent process flow supported by the new system is described from the perspective of nursing homes and insurance associations as initiators.

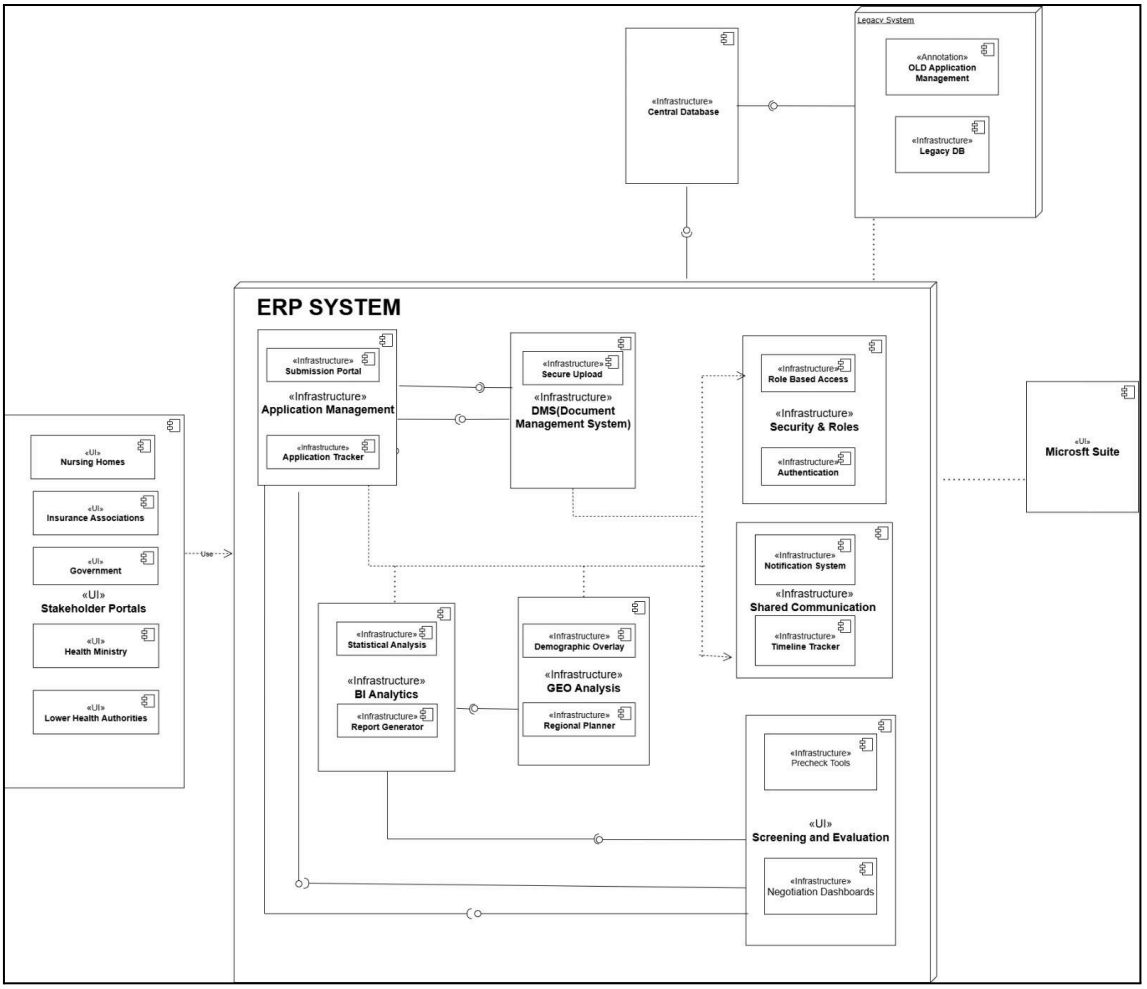
Figure 3.3 UML use case diagram



Within the new system, the planning procedure starts with the application submission by the initiators. The system will provide a structured entry point for the initiation, capturing all required data and ensuring that the process is properly logged from the start. This action replaces the current manual process of sending documents via email by centralizing the submission and thus providing the traceability of the application. Tracking application status lets all relevant stakeholders view the real-time progress of the procedure in the status page. This way, the system allows the health ministry and relevant government agencies to be more actively engaged in the process, moving away from their current passive role. Furthermore, approving and rejecting applications are embedded functions in the system, allowing it to automatically notify relevant stakeholders of any updates. The digitized screening process replaces the manual reviews and communications and, in case of initial approval by the government agency, can notify the relevant actors to start the negotiations. Negotiations between nursing homes and insurance associations are primarily held outside of the system; however, the system still supports the communication and document exchange between parties. Evaluation of negotiations, which involves conducting various analyses, is supported by the integration of BI tools, replacing the use of outdated and disparate tools in the evaluation process. After reviewing the generated report, the health ministry has made the decision and issued a decree; the government uses the system to issue the decision notice. Then the system ensures it is documented, distributed to all stakeholders, and its data is stored in a centralized database for future reference. This action finalizes the process and marks the completion of the procedure.

Phase C – Application Architecture: This section has the core components of the backend that power the new nursing home system for nursing home planning. The overall architecture of the system is shown in Figure 3.4 of the UML component diagram. Each module has been thoughtfully designed to fix real gaps that were observed in this case, including outdated tools, scattered communication, and a lack of integration across stakeholders. The overall architecture follows TOGAF ADM's Application and Technology layers, ensuring a modular, scalable, and interoperable system (Ji et al., 2022, p. 2).

Figure 3.4 UML Component Diagram (Version 1)



This component diagram covers the backend design of a modular based system, where there is a clear structure on how the artifact will be modularised to meet the specific requirements. Following Kashyap et al. (2006) this figure shows a clean architectural design visualises dependencies and how the system functions.

Stakeholder Portals - User Interfaces (UI): This module serves as a crucial bridge between everyday users and the backend logic running behind the scenes. Thus, the UI facilitates stakeholders' interaction with the system. Through this interface, the actors will submit documents, track progress, manage negotiations, and access communication tools. This layer serves as the crucial bridge between everyday users and the backend logic running behind the scenes.

Application Management Module: This module facilitates the complete journey of a nursing home application, from submission to final review. It is mostly used by care providers when entering application details, uploading structured documents including

staffing plans and architectural layouts, and tracking their status in real time. Once submitted, the data is passed to the Document Management System (DMS) and stored in the central database. It will also be connected and parsing information through to the Screening and Evaluation module.

Document Management System (DMS): The DMS keeps all transactions transparent and auditable (Jordan et al., 2022, p. 8). It securely stores all documents, tracks versions, and adds metadata for easy traceability. Every file gets time-stamped and tied to a stakeholder identity using role-based access control. While the DMS works independently, it works with the central database for indexing and document retrieval.

Security and Role-Based Access Control (RBAC): This module enhances system security and protects government information resources (Chuanfan, 2010, p. 1). Using the RBAC model, it manages which modules are accessible to which actors, who can access what, based on their role. Security features are integrated into the design of the system from the beginning (Royer, 2013, pp.51–54). Administrators get dedicated interfaces to create and adjust roles, keeping things flexible yet fully compliant with privacy regulations. i.e., General Data Protection Regulation(GDPR)(see Section 3.3.2).

Screening and Evaluation Module: This module automates how applications are assessed. For instance, some use cases such as looking at staffing plans, care quotas, and local demographics. This module will receive data and information from the Application module and have access to the Databases and DMS modules for decision making. It makes the review process more efficient and helps evaluators to make evidence-driven decisions by providing calculations, visual feedback, and decision-support tools.

Shared Communication and Notification Services: This component manages all communication functionalities, including automated reminders, status updates, negotiation notifications, and related alerts. It ensures that all stakeholders remain informed throughout the process. It sends alert messages regarding missing documents, approaching deadlines, or ongoing discussions. Additionally, it maintains a detailed record of correspondence during negotiation phases, promoting transparency and accountability among all involved actors.

Integration with External Tools: To keep the system adaptable and scalable, the system is built with the ability to connect with third-party tools as needed. These third-party tools include geospatial analysis tools, BI platforms, and Microsoft Suite

Compatibility. Geospatial analysis tools are used for mapping care capacity and spotting regional demand trends. This integration can provide a more comprehensive view of business operations by incorporating geographic context into decision-making processes (Sternad Zabukovšek et al., 2022, p. 24). BI platforms generate data-driven reports and identify long-term planning patterns. Thus, BI tools enhance ERP systems by offering capabilities such as data mining, OLAP (Online Analytical Processing), reporting, and query functionalities (Nofal & Yusof, 2013, p. 4). Then, Microsoft Suite Compatibility ensures users can still work with legacy formats such as Word and Excel during the transition phase. Public sector organizations must thoroughly evaluate the long-term impacts of maintaining and operating legacy systems, as well as determining the right time to retire them, all within a structured change management and project planning framework (Irani et al., 2023, p. 11). To ensure the legacy system integration in this case, the proposed solution is designed to connect with existing workflows during a phased rollout. This way, legacy tools will continue to feed data into the system but will not receive updates, thus allowing for a smooth, low-risk migration to the new platform without disruption. Together, these backend components form the digital backbone of the ERP-BI system. Built around TOGAF ADM's architecture layers, the design ensures flexibility, traceability, and seamless collaboration across sectors, and all in real time (Ovalle et al., 2024, p. 3).

Phase D – Technology Architecture: This phase specifies the underlying infrastructure required to support the Digital Nursing Home solution. This includes servers, hosting environments, data storage systems, and external integration points. The Centralized Database is the system's main data vault. It holds everything from application forms and negotiation results to evaluation reports and dashboard data. It is designed for high availability and structured querying, giving stakeholders fast access to what they need (Mohammed & Saleh, 2017, p. 2). Unlike the fragmented systems in the current process this unified setup prevents data silos and supports cross-functional insight. While detailed technical implementation is beyond the project's scope, the system is envisioned to run on a secure cloud platform with centralized database architecture. APIs enable integration with external tools such as Geographic Information System (GIS) platforms and the existing legacy system. Role-based access control and compliance with GDPR are embedded in the infrastructure layer.

Phase E – Opportunities and Solutions: The modular Nursing Home System design allows each region or stakeholder group to adopt components gradually. Opportunities include integration with existing demographic datasets and visual BI tools, and leveraging Microsoft Suite compatibility to bridge current workflows. The combination of BI with ERP ensures both operational management and long-term strategic planning can be conducted within one system (Moussas et al., 2024, p. 11).

Phase F – Migration Planning: Migration from legacy tools (such as spreadsheets and email-based submission) is planned as a phased process. A structured migration planning strategy is employed to transition nursing homes and affiliated stakeholders from fragmented legacy systems such as spreadsheets, email-based communication, and isolated databases into a centralized Nursing Home System. The ERP migration process often involves a series of integration challenges, such as ensuring compatibility with legacy systems, addressing complexities in data migration, and securing buy-in and effective adoption from end-users (Sreenivasa Rao Sola, 2021, p. 2). This phased process includes digitizing existing documents via a Document Management System (DMS) and integrating critical historical data using hybrid approaches, such as data cleaning, standardization, and ETL (Extract Transform Load) pipelines. ETL pipelines are essential for migrating structured and semi-structured data from legacy sources into the ERP system while preserving referential integrity (Singu, 2022, p. 3). Legacy systems remain accessible in read-only mode (Bing Wu et al., n.d., p. 7) to preserve institutional memory, while API or semi-manual bridges address non-automatable records. Operating legacy and new systems in parallel allows for gradual rollout, minimizing risks and ensuring stability . Ultimately, this migration empowers real-time business intelligence analysis by consolidating historical and operational data, supporting policy continuity and strategic decision-making. With a clear and thoughtful migration plan, important historical data stays reliable and easy to access in the new nursing home system. This helps teams make decisions based on real-time insights while keeping a strong connection to past records. The transition is transparent and follows the rules and standards expected in public sector environments.

Phase G – Implementation Governance: Although full deployment is beyond the project's scope, a governance mechanism is proposed via stakeholder walkthroughs, prototyping, and expert evaluations. These ensure that real-world processes are reflected accurately and that system usability meets the expectations of all involved actors.

Phase H – Architecture Change Management: The Nursing Home system is designed with adaptability in mind. Policy changes, role shifts, or new planning criteria can be accommodated by modifying individual modules without altering the entire architecture. This ensures long-term sustainability of the system and compliance with evolving regulatory or demographic contexts.

3.3.2. Rights and Roles (RBAC)

Role-based access control (RBAC) model is a proven methodology for large-scale authorization management that assigns permissions to users based on their organizational roles rather than individual identities (Ferraiolo et al., 2001, pp. 1, 3). The National Institute of Standards and Technology (NIST) standard defines three fundamental RBAC models: core, hierarchical, and constrained RBAC models (Ferraiolo et al., 2001; R. Sandhu et al., 2000; R. S. Sandhu et al., 1996). Core RBAC model, also referred to as flat or base RBAC model, provides the foundation with basic user-role permission assignments, while hierarchical RBAC introduces role inheritance that mirrors organizational structures. However, the constrained model emerges as the most sophisticated model due to its incorporation of separation of duties constraints that prevent conflicts of interest and ensure regulatory compliance (Ferraiolo et al., 2001, pp. 15–17; R. S. Sandhu et al., 1996, p. 7). The model implements both static separation of duties (SSD) relations, which puts constraints on the assignments of users to roles to ensure no user gains conflicting permissions (Ferraiolo et al., 2001, p. 15). Additionally, dynamic separation of duties (DSD) relations, which is similar to SSD however "limits the availability of permissions by placing constraints on the roles that can be activated within or across user's sessions" (Ferraiolo et al., 2001, p. 17).

The nursing home planning process involves multiple stakeholders with distinct responsibilities and access requirements. This multi-stakeholder environment requires a sophisticated access control that can address complex organizational structures, regulatory requirements and dynamic security needs which are present in the nursing home planning process. Therefore, the proposed system implements a constrained RBAC model with both SSD and DSD, and decentralised RBAC management. Core implementation components include user-role assignment functions that validate SSD constraints, session management that enforces DSD limitations, and administrative

RBAC management review functions that enable continuous monitoring of constraints compliance.

The main SSD constraint in the system is between initiation and decision making, which is to prevent the same actor from both initiating and making final approval decisions. As a result, nursing homes and insurance associations can not be assigned to the Health Ministry or government agencies' decision-making roles. DSD constraints include phased-based role activation, which refers to the possibility of permissions change in certain phases; and institution-specific access data, referring to the fact that users can only access data from their own institution during active sessions. As seen in Figure 3.5, the government is a system administrator, however the top state government is a senior system administrator who gives lower level state authorities manager permissions to add, delete and edit users and give them permissions. This emphasizes the high power and interest of top state governments within the system in accordance with Eden and Ackermann (1998, p.122).

Figure 3.5 RBAC matrix

Users	**Roles**	**Permissions**	
		Read	**Write**
Health Ministry	Decision-maker	All	Decisions
	Evaluator	All	Decisions
Government	Administrator	All	-
	Initiator	Own data	Proposal Submissions
	Evaluator	Negotiation documents	Reports, Decisions
	Decision maker	All	Decisions
Insurance Association	Initiator	Own data, Decisions	Proposal Submissions
	Negotiator	Own data, Decisions	Comments
Nursing Home	Initiator	Own data, Decisions	Proposal

			Submissions
	Negotiator	Own data, Decisions	Comments
Lower level health authorities	Evaluator	Negotiation documents, Reports	Comments

3.3.3. User Interface

To ensure better and easy understanding of the system architecture, the design of a governmental web-portal on the basis of a proposed system was designed. Sandoval-Almazan & Gil-Garcia (2010) argue that an administrative portal must carry out certain characteristics, including a process-oriented website, interoperability, modularity and flexibility, compliance and traceability, role-based access and security, and standardization and user-centricity (pp. 72-81). This website digitalizes the existing process instead of reinventing the new model as the digitalization approach in Germany is rather traditional instead of radical, digitalizing what has been there. (Sommer & Proff, 2023, p. 6).

Figure 3.6 Log In page

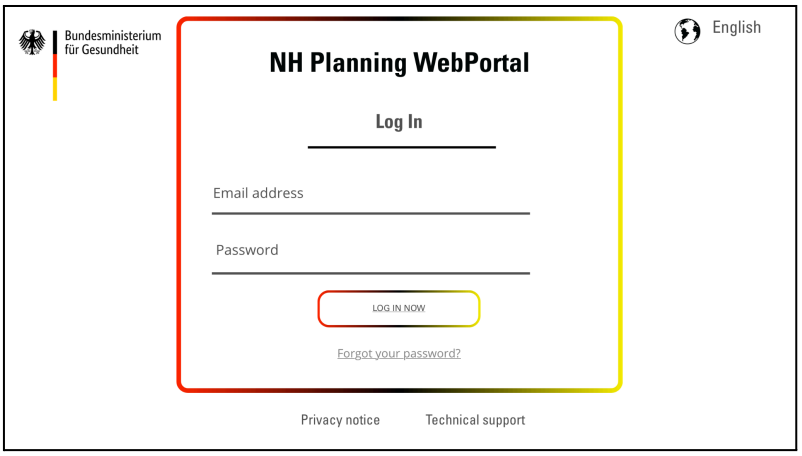
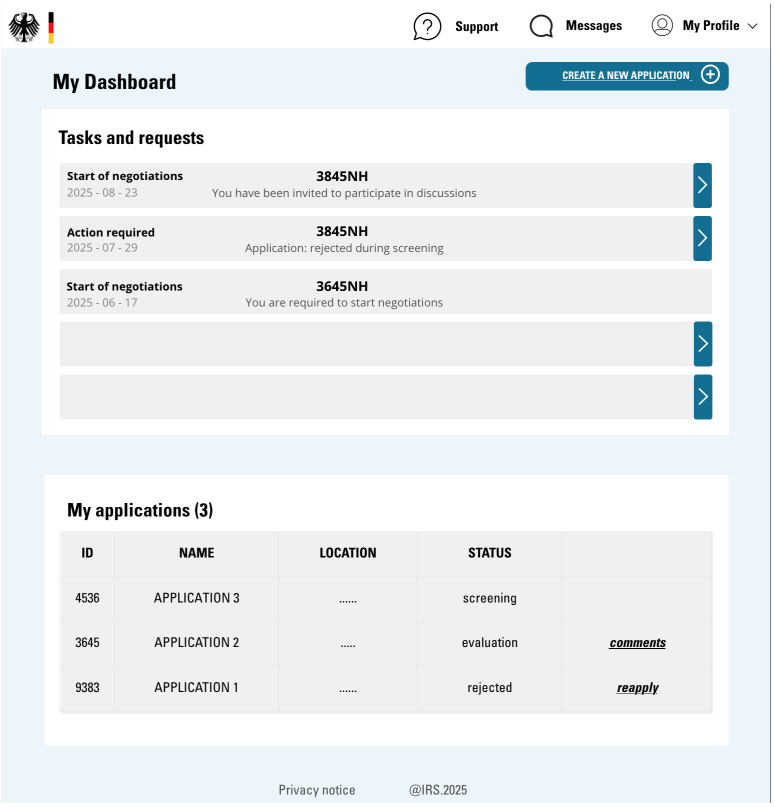


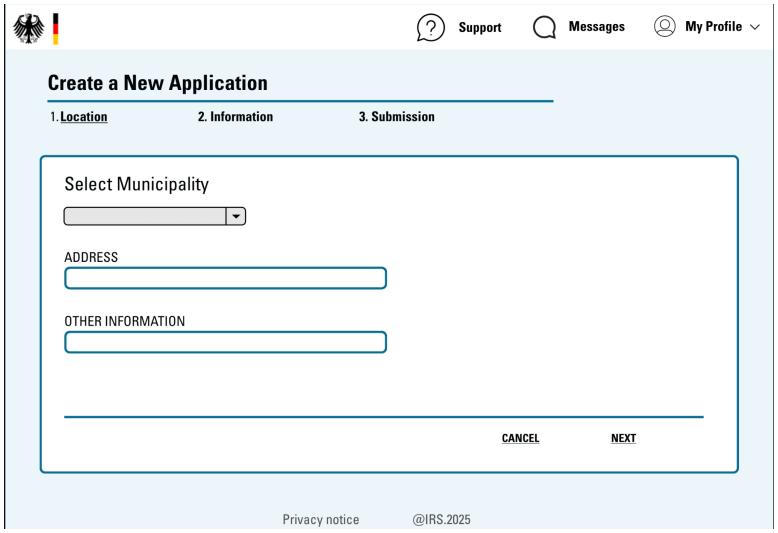
Figure 3.6 portrays the *log-in page*, providing secure access to stakeholders by providing password requirements, routing to role-based authentication, and a 'forgot password' option. Clicking the log-in button conveys stakeholders to the My Dashboard page (Figure 3.7).

Figure 3.7 Nursing home dashboard



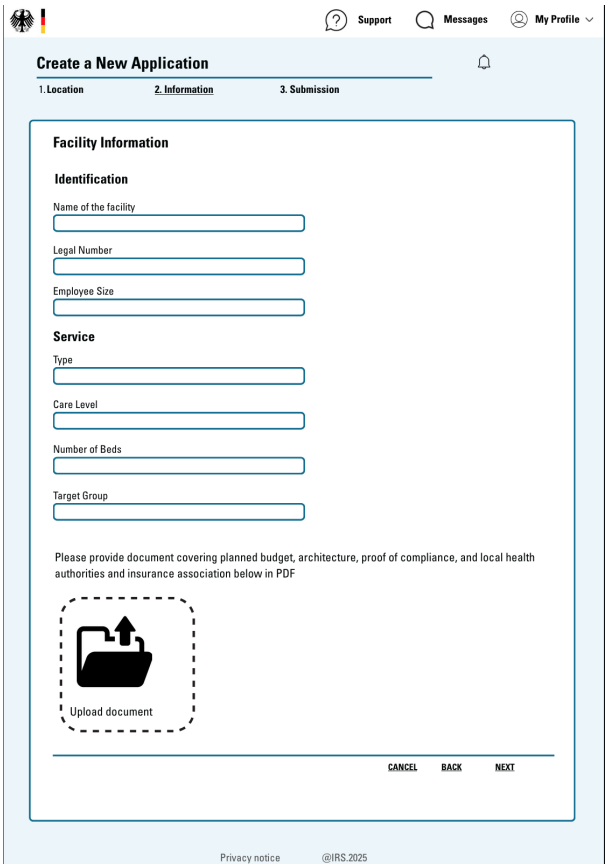
"*My dashboard*" page conveys an overview of tasks and requests made by other users in the system. As seen in Figure 3.7 the button "create a new application" leads to the application opening, which consists of three steps. In the first step, the nursing home can select the municipality and detail the address in the given boxes (Figure 3.8 (a)). This page presents an information-driven approach, one of the website's functions, to elevate transparency in the system.

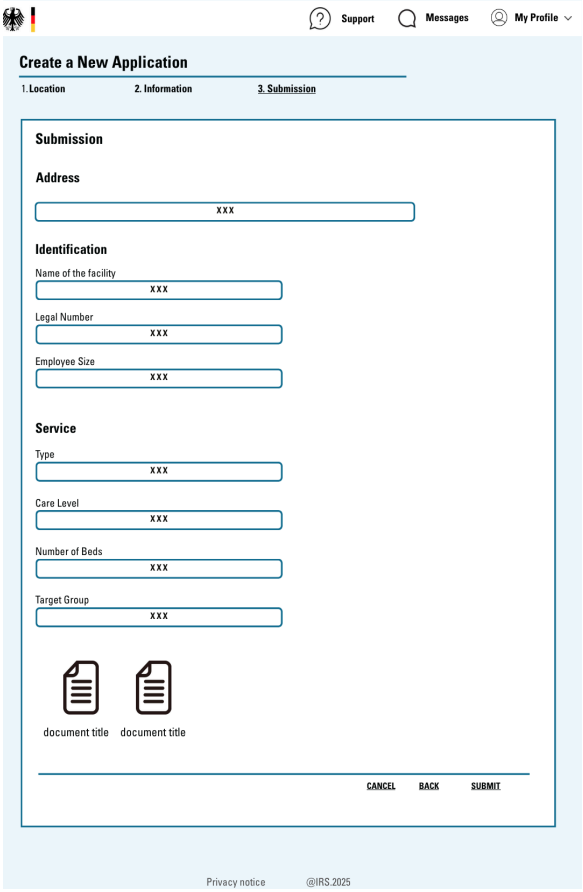
Figure 3.8(a) Step 1 of a new application nursing home creation



This website offers flexibility, serving exploration and orientation sense-making criteria by Krug (2014), as the bottom section provides the option to go to the next step with 'next' or allows the user to return to the previous page using the 'back' button. The 'back' button in regards to the breadcrumb theory of websites does not solely provide context awareness and fast decision-making but clarifies the navigation (Krug, 2014, p. 89). Once 'next' is clicked, it goes to the next step (Figure 3.8(b)), which is the detailed input of nursing home submission. The last step is submission (Figure 3.8(c)), in which the website offers the summary of the application to review before submission.

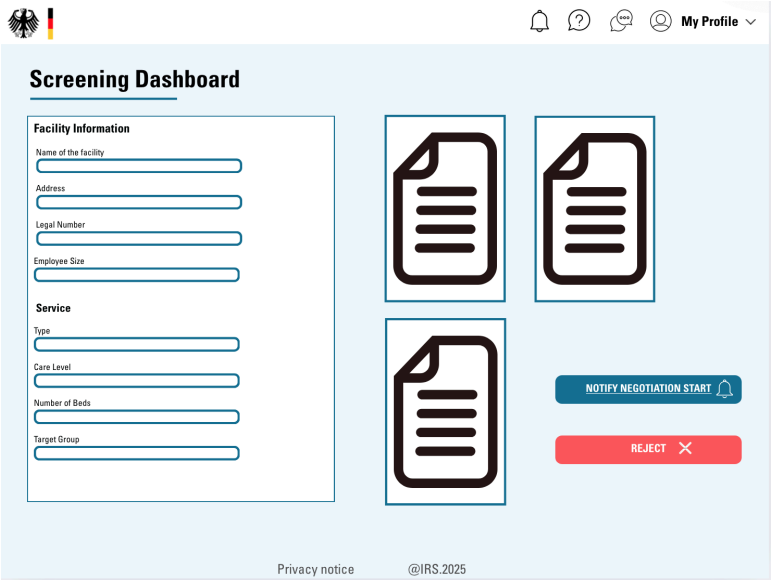
Figure 3.8(b) and (c) Step 2 and 3 of a new application nursing home creation





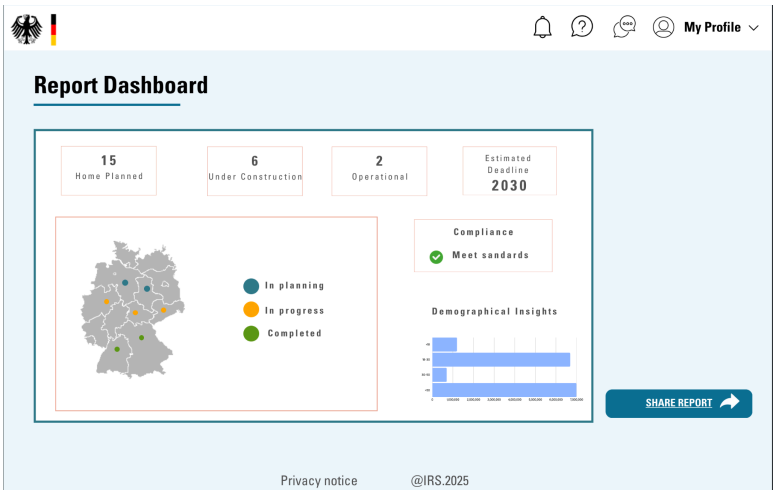
The website features a screening dashboard (Figure 3.9) and a report dashboard (Figure 3.10); the former can be monitored by the government and the health ministry only to fulfill the function of both bodies. Whereas the latter would be accessible merely to the government to carry out reporting and decision processes.

Figure 3.9 Screening dashboard for government



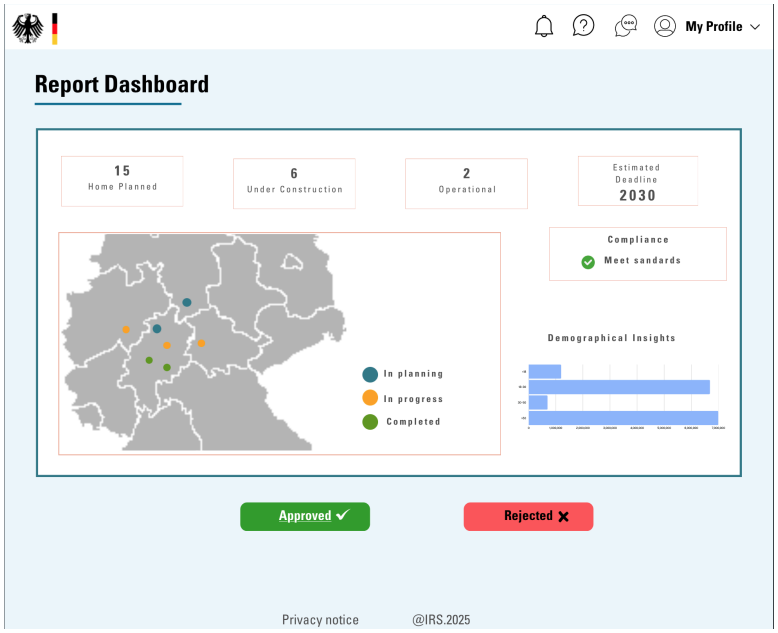
PDF signs enable users to locate the document submission space. This does not only reduce cognitive load but also orientation to the task or documents needed in the page.

Figure 3.10 Report creation dashboard of the government



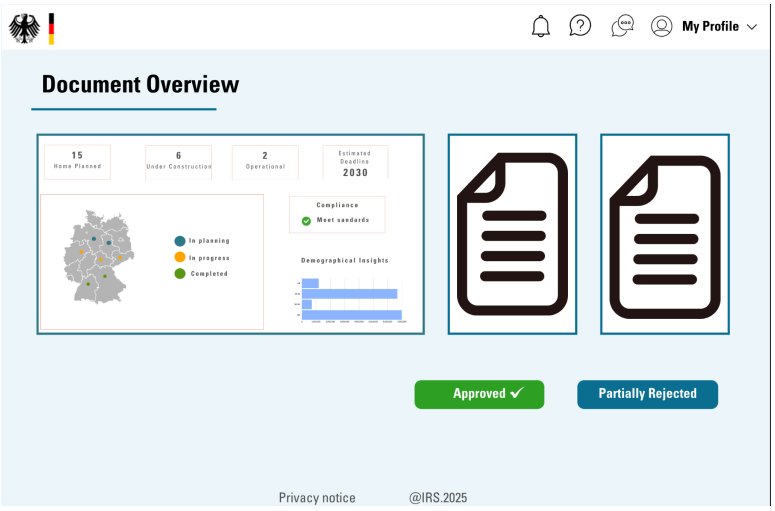
The report created by the government can further be shared with the health ministry to make the decision whether to approve or disapprove (Figure 3.11).

Figure 3.11 Report dashboard for the health ministry



The document overview is for the decision notice as the last step of the nursing home planning system for the government to notify the final result of the decision (Figure 3.12). Following the planning system procedure, three decision options which are approved, partially rejected, or rejected are labeled in button visuals to create a self-explanatory button in accordance with Krug's approach. However, Norman (2002) emphasized that affordance perception is dynamic, as past experiences of individuals can interfere with perception of things (p. 19).

Figure 3.12 Document overview



3.4. Activity 4 - Demonstration

Stakeholder walkthroughs were conducted to demonstrate the functionality and applicability of the proposed nursing home planning platform. Both components, conceptual system architecture and front-end, of the developed artifact were used during the demonstration. The goal in this activity was to show how the proposed solution addresses the identified weaknesses in the current nursing home planning process. Each walkthrough was tailored to the stakeholder's area of expertise. To Respondent 1 the conceptual system architecture was demonstrated , and to Respondent 2 - the UI prototypes. Respondents' information is shown in Figure 3.13. The walkthroughs focused on both the process flow and the functionality mapping to help participants understand how the system's modules and interfaces would function in practice.

Figure 3.13 Informants' table

Informant	***Position***	***Organization***	***Interview Duration***
Informant A	Federal IT Architecture Management	FITKO	1 hour, online
Informant B	*Pflegehilfskraft*	Deutsches Rotes Kreuz (DRK)	30 minutes, online
Informant C	*Senior Software Engineer*	BBD Software	written

3.5. Activity 5 - Evaluation

The evaluation of the artifact was conducted by semi-structured interviews with the same three respondents. Through the evaluation, the research aimed to validate that the conceptual model addresses real-world needs and reflects stakeholder workflows. Evaluation criteria included (i) the system's technical soundness, (ii) feasibility and scalability of the proposed system architecture, (iii) alignment with legal and organizational requirements, and (iv) ease of use for UI. Respondents were asked both

predefined and follow-up questions to examine strengths and weaknesses of the solution, suggestions for improvement, and potential barriers to adoption.

3.6. Activity 6 - Communication

The results and the conceptual artifact of the research was communicated through presentations and expressed in a written research report. Additionally, during the research and development of the artifact, two presentations were held by the researchers. First, an informative presentation was given to discuss the main identified problem and a solution proposal. The second presentation—a pitch delivered near the end of the development—was met by great acclaim and acceptance. The main beneficiaries of this research are state governments, which wish to implement such systems to digitalize the nursing home planning process in a collaborative manner.

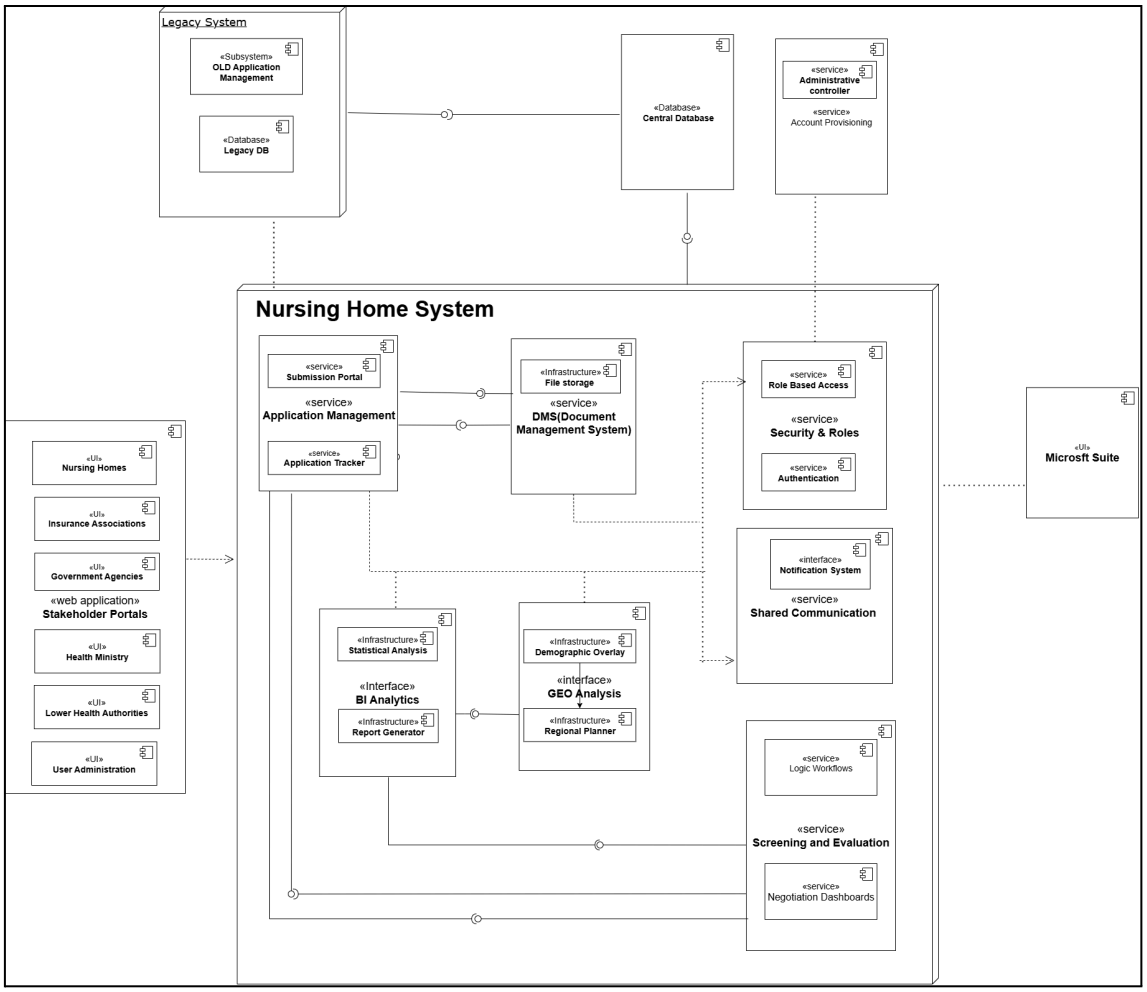
4. Results

To assess and improve the initial system and UI design for the nursing home planning, semi-structured evaluation interviews were conducted. The purpose of the evaluation was to critically analyze whether the proposed solution design is technically sound, feasible, scalable, covers the planning process, and aligns with legal and organizational requirements.

Respondent A as an architecture expert identified significant issues with unclear terminology and ambiguous concept definitions in both system explanations and diagrams. For instance, the “evaluation and negotiation module” lacked clarity in its functions, whether it entailed simple documentation handling and manual decisions or more complex, automated assessments. Furthermore, concerns over the storage of relational and non-relational data arose. Most importantly, Respondent A stressed the importance of robust identity verification, and suggested the integration with external national identity (country-wide) providers, especially when it concerns institutional users. These insights led to concrete improvements in the system architecture, such as clarity in terminology, refinement of existing module designs, and addition of new modules into the system.

Built on the ground of software thinking and expert feedback from TOGAF's view, The design process was developed iteratively, firstly designed directly from the case and thereafter administered to experts for evaluation. The proposed architecture design aims to enable structured application with tracking capabilities for better management and traceability. Supporting both manual and automated evaluation procedures from multiple stakeholders and facilitating the negotiations (back and forth) that take place. This is all to ensure transparency and auditability of decisions while adhering to legal and regulatory standards found in the SGB (*Pflegeversicherung*, n.d.; Mergel, I. 2021). The first iteration of the artifact was a design of a Component Diagram, highlighting each component that is part of the nursing home project as shown in Figure 3.5 UML Component Diagram (Version 1). Due to numerous misalignments shown in the initial design in comparison to the intended purpose of the system, after interviews and alignment though evaluation, this modal was created.

Figure 4.1 UML Component Diagram (Version 2)



To address any form of management security, the design was changed to incorporate an administration control user interface. This component will be accessed by Administrator accounts only, from respective stakeholder organizations with the sole purpose of managing accounts in the form of creating, adjusting or removing accounts of users. Not only does this feature support security (Royer, D, 2013 p. 27) it was also a direct suggestion from the reviewer (Respondent A). This has improved flexibility within the security of the system by substituting an Active Directory (AD) security framework to an Identity and Access Management(IAM) design which is designed to accommodate federated providers (BundID, enterprise SSO) while supporting standard protocols (OAuth2, SAML).

Other key considerations that were taken into account were the incorporation of Microsoft as a module in the design. While in the initial iteration it played a major role in security and as an external tool for communication. While the AD approach is canceled, it does support smooth onboarding and gradual change management if it can

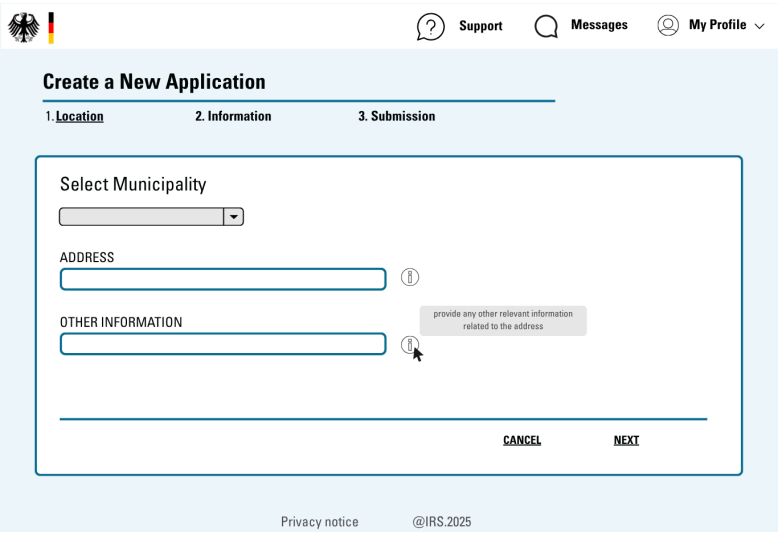
be integrated for communication. Ideally each user of the system will have a user account that is linked to one government agency's domain.

From an online review of the original design by a software engineer expert (Respondent C) the feedback highlighted the importance of a web-based modular approach for such a complex system with multiple stakeholders. This feedback encouraged the designers to keep the modularity design and create a User Interface design to make review of the artifact easier for non-technical stakeholders. For governmental use it was important to identify design that would make it easily relatable to users with less technical backgrounds as well as showing familiarity to other available public systems (Buko, 2018 p. 27-56). Without extensive knowledge on German public systems the team proposed minimalistic material design elements to make it intuitive and professional (Ko et al. 2018, p.5). The colour scheme chosen was primarily a blue and white monochromatic palette.

The interview with Informant B inspires the second iteration to have a 'tooltip,' as the informant believes that such a feature would be helpful and is common on German websites. Additionally, In response, a context-sensitive tooltip feature was introduced to the application form interface (Figure 4.2). This feature provides users with immediate explanations for each field form. It is activated automatically when the user hovers the mouse over this tooltip (Figure 4.3).

Thus, the UI now better supports users in accurately completing the application process. The informant also emphasizes that the back button is extremely helpful, which validates Krug's proposal on the practicality of website sensemaking.

Figure 4.2 The application form with tooltip

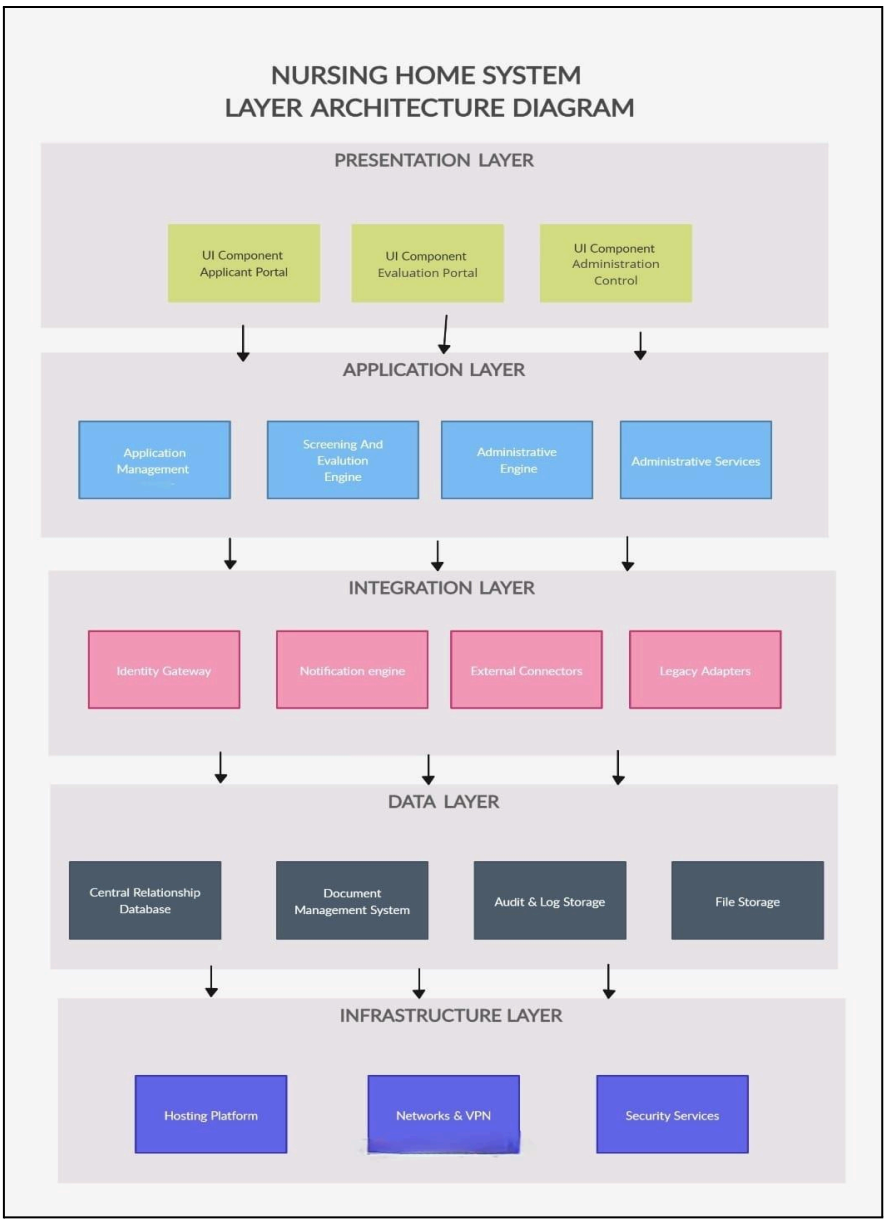


A layered architecture diagram (Figure 4.3) was later developed in accordance with the system to reduce the complexity of the application by clearly separating each layer's responsibilities. Dividing each layer shows more detail to the implementation of TOGAF within the design process of the solution. These layers emphasize the importance of separations of concerns and responsibilities. Joshi (2016, p.23) and Vo, (2015) mention the importance of looking at the architecture in isolated layers for modularity and scalability. This sort of representation will assist with stakeholder evaluation in coming iterations as well as testability in implementation. The layer is made to make the connection visible between modules in a complex system akin to the Nursing Home System. This enables functionalities or benefits focusing on application, and technology, ensuring modular, scalable, and interoperable methods. The architecture consists of the following components:

User Interface Layer: The Applicant Portal will be utilized by care providers, the *Evaluator Portal* used by insurance associations and health ministries, and the *Admin Console* , used by system administrators for managing roles, users, and system configurations. These interfaces will facilitate all user processes, such as application submission, progress tracking, document upload, and stakeholder communication.

Application Service Layer: This layer includes the *Application Management Module* which manages the lifecycle of nursing home applications from submission to decision. It also includes functionality for uploading structured documents (staffing plans, building layouts), tracking application status, and initiating evaluations. The *Screening and Evaluation Engine* which conducts rule-based checks based on care quotas, demographic alignment, and staffing levels. Checks are based on parameters aligned with §1 and §72 SGB XI. It also provides contextual decision support for human reviewers, for human-in-the-loop (Güner et al., 2022 p. 170-174). *Negotiation Support Module* records and enables users to create dispute and resolution during the review process. Negotiations can occur inside or outside the system. This is to ensure transparency of outcomes. *Administration Service* enables the technical staff to manage roles, identities, module access, and configuration settings as advised by Respondent A, similar to Royer, 2013.

Figure 4.3 Layered Architecture Model



Application Service Layer: This layer includes the *Application Management Module* which manages the lifecycle of nursing home applications from submission to decision. It also includes functionality for uploading structured documents (staffing plans, building layouts), tracking application status, and initiating evaluations. The *Screening and Evaluation Engine,* which conducts rule-based checks based on care quotas, demographic alignment, and staffing levels. Checks are based on parameters aligned with §1 and §72 SGB XI. It also provides contextual decision support for human reviewers, for human-in-the-loop (Güner et al., 2022 p. 170-174). *Negotiation Support Module* records and enables users to create dispute and resolution during the

review process. Negotiations can occur inside or outside the system. This is to ensure transparency of outcomes. *Administration Service* enables the technical staff to manage roles, identities, module access, and configuration settings as advised by Respondent A, similar to Royer, 2013.

Integration Layer: This layer will be handling all essential connections to the main system (Harris et al., 2008). An Identity Gateway is needed to integrate with external Identity and Access Management (IAM) providers such as BundID or others, ensuring that only verified actors can access sensitive modules. Notification Engine is envisioned to manage alerts and messages such as reminders and status messages across stakeholders. Other tools will be applicable for our use cases, and therefore *External Tool Connectors* will be a module to ensure interoperability with Geospatial Analysis tools (for demand mapping), Business Intelligence platforms (for statistical, descriptive, and geospatial analysis and other data functionalities), and Microsoft Office compatibility for backward integration with Excel/Word formats. Legacy Adapter modules bridge existing legacy systems during a phased rollout, which is to support one-way data syncing without maintaining bidirectional dependencies. However, it will also work hand in hand with other dependencies from existing tools in the legacy system (Harris et al., 2008 p. 45–51).

Data Layer: central database stores structured application data, documents, evaluation metrics, and decision records. While the Document Management System (DMS) securely stores uploaded files, supports versioning, and applies metadata and timestamping. Linked to role-based access control policies. The file storage layer preserves long-term files for archiving, compliant with retention and legal regulations. An Audit & Log Storage module captures all system-level engagements, communications,and traceability for compliance audits.

Infrastructure & Platform Layer: The system needs to be hosted in a way that supports deployment on a secure government-operated or certified European cloud infrastructure that is handled by Hosting Platform. Taking into account Networking & VPNs. Accounts for Germany's federal-municipal networking constraints (e.g., DOI VPN). Security Services implements TLS encryption, secure endpoint policies, and IAM enforcement.

5. Discussion

5.1. Theoretical and Practical Implication

The development of the new nursing home system contributed to several key theoretical areas. A primary contribution lies in applying the TOGAF framework to the new planning system, showing how enterprise architecture can bridge the gap between private sector efficiency models and public sector governance. This approach helps to operationalize NPM principles, indicating the demand of transparency and agility of government autonomy when digitally pushed. By embedding both stakeholder and legal requirements, it also addresses the common problem of fragmented systems, which was one of the core problems in this study. Additionally, having TOGAF helps integrate business views into public sector cases, potentially unlocking business-oriented benefits aside from public value. Besides, the efficiency provided by TOGAF suits streamlining processes in a low-cost manner given the nature of public organizations that are far from being profit-oriented. Furthermore, the user interfaces' design guided by sense-making theory ensures the web-portal meets the needs of diverse users to increase adoption, trust and transparency. This also highlights the need of digital government in both transparency and agility. RBAC is a core theoretical component to ensure security and separation of duties in the new proposed system. It supports a "compliance-by-design" approach, implying that the regulatory context must be integrated intrinsically instead of embracement in a post-hoc process. This strengthens the argument that creating a public value in digital transformation requires legal compliance, stakeholder alignment, and service efficiency altogether.

This research pursues high feasibility by considering the practicality of approaches selected. This research pursues high feasibility by considering the practicality of approaches selected. From a practical standpoint, ERP is highly feasible for its integration with legacy systems and processes. Therefore, it makes the research artifact applicable for integration into existing systems and to health care planning issues in Germany. By providing real-time tracking, transparency, clear roles definition, and focuses on legal compliance and system security to support GDPR requirements. Hence, this will elaborate on multi-faceted factors of the regulatory approach, governance, and accountability in the nursing home planning system's digitalization to create equity.

5.2. Legal Implication

The design of a digital platform for nursing home planning can not be viewed solely as a technical path but rather as a multi-layered context of legal obligations and governance structures. From the political landscape, Germany is path-dependent on the rule of law in terms of digitalizing public service (Sommer & Proff, 2023. p. 6). Hence, migrating old data from legacy systems must be considered. Deriving from §§72, 75, 112–114 of SGB XI and §69–71 of SGB V, RBAC is guided to pursue duties of stakeholders and reduce ambiguity of the planning system to create minimum conflict system usage and design (Ferraiolo et al., 2001; Sandhu et al., 1996).

The informant A stated that the regulation translation into the system remains vague, which has triggered more or less unsubstantiated regulatory practice. These findings emphasize that socio-technical aspects need to be defined and intertwined with legal aspects. This motivates the integration of Public-Private Partnership (PPP) into the ERP or digital transformation. Thus, it calls for enforcing a more regulatory approach accommodating Public Private Partnership (PPP) or participatory governance, echoing the argument by Kolehmainen et al. (2024) and Stoker (1998). Instead of focusing on formal regulation, informal regulation that construes individual aspects of participation, as in social values and culture, must be reckoned with. Li et al. (2024) enforces that informal and formal regulations can amplify environmental, sustainability, and governance effects of digital transformation, with informal regulation having the higher impact.

5.3. Design Implication: Alternatives Analysis on Standard Software or Custom Development

This research aims to design architecture that centralizes data to reduce data silos with affordable options. As a result, this system does not only provide real-time visibility but also consistent information flows, supporting cross-functional insight as well. Thus, the proposed system within the research optimizes current weaknesses of fragmentation surrounding the current system. This presents as the supportive tool of the current system.

To achieve a more efficient process, this research intends to embrace ERP as it is the most feasible and familiar tool in Germany. Since most standard ERP systems are designed specifically for regular businesses, customization by ERP systems, using

modules and middleware, provides more flexibility for public sector organizations by providing better integration capabilities across different systems, compliance and adaptation to regulatory changes, and resilience in the long term. (Hustad & Stensholt, 2023, p. 2). Hence, in the decision to integrate standard and custom-made software, in the research the custom-made was selected as the prominent one.

There are four subprocesses that become the main considerations, including (1) application submission and receipt, (2) process management and execution, (3) stakeholder coordination and reporting, and (4) data-centric evaluation and analysis. ERP handles connection, allows scalability, is relatively stable and compliant with the law (Hustad & Stensholt, 2023, p. 2). Some tools, including Salesforce Public Sector or Microsoft Dynamics are basic and do not fit the given regulation, including §72 SGB XI. This external condition is posed by the lack of third-party integration (Sinard & Gershkovich, 2012, p. 1). This implies the need for a customized application submission that has a legal assessment and is flexible but possible to be integrated with the document management module.

Coordinating stakeholders and their reporting is likely to be facilitated by complex tools, including SharePoint or DocuWare, which might delay the adoption in the public sector with technocratic issues and can lead to greater problems with resistance to change (Khumalo & Mearns, 2019, p. 1). A Document Management System (DMS) equipped with audit trails and metadata is necessary for keeping transparency and meeting the law requirements. Customized ERP allows easy switching from old systems and adds support for old data types and document flows (Murthy, 2021, p. 1).

Report generation process management system (DMS) has potential 'satisficing' solutions for showing data patterns, population needs, and where gaps in care exist (Parthe, 2023, p. 3). These tools are well suited for this subprocess and can be integrated with the ERP's centralized database and Screening & Evaluation module. Therefore, for this subprocess, market software is recommended. BI systems must offer advanced visualization techniques and enable automatic reporting for authorities. Important things to include: a real-time dashboard for Nursing Home needs, a score system based on people data, workers, and what is available. It is equipped with visual aid found heatmaps, and reports' exporting features to PDF or Excel. Organizations that use BI

tools well can handle today's healthcare challenges more easily and stay ahead of the competition by making smart, data-driven decisions (Devika Rajhamundry, 2024, p. 7).

In digital public systems, flexibility remains vital. Alternative platforms such as Mendix, OutSystems, or Microsoft Power Platform give faster ways to build tools with less coding (Shi et al., 2025, p. 1). These help to test ideas, show progress to people, and quickly adapt to changes in regulations while still adhering to GDPR. Although ERP inspired the whole proposed design, the research acknowledges some low-code alternatives that merit attention but remain traceable and flexible. In terms of qualified BI, reporting can be made efficient with Microsoft Power BI, QlikView, and Tableau, employing OLAP, which offers mature solutions for visualizing trends, population needs, and capacity gaps (Parthe, 2023, p. 3). These tools are well suited for this subprocess and can be seamlessly embedded into the ERP's centralized database and Screening & Evaluation module. Therefore, for this subprocess, market software is fit for purpose.

Thus, custom-made seems to fix the gaps in an organization when there is a mismatch between the functionalities of a standard ERP and the specific needs of the organization (Ng, 2013, p. 6). Informant 'A' strongly suggested open-source demonstrated Pega that delivers basic infrastructure for this system. In comparison to Camunda and Pega offer simple and low-code customization, supporting workflow automation with rapid UI design (Khumalo & Mearns, 2019, p. 1). This allows low-code adjustments that respond to possible changes to external factors, as in regulatory changes and aligning different interests for the general public (Shi et al., 2025, p. 1; Murthy, 2021, p. 1).

The dynamic external factor affecting the public sector elicits agility. Low-code remains applicable in catalyzing digital transformation. In addition to that, a blended approach, combining low-code, no-code, and conventional coding approaches, can accelerate rollout but keep the implementation sustainable and profitable in the long run (Ajiboye, 2021, p. 5). Hence, the alternatives must be context-dependent and adjusted to resources available within the public sector organization. The approach streamlines modernizing the current system without changing any single process or legacy system.

6. Conclusion

The overarching problem that this research addresses is the fragmentation of the nursing home planning system. Design science research is ultimately not concerned with finding the best approach. Rather, it seeks to support practitioners in their daily work by improving what they already do. In this way it directly offers what descriptive or exploratory research can only allude to in the long term (Weggeman & Cauffman, 2024, 8). Accounting for Germany's domination of ERP, the complex system is integrated into ERP, enriching the research field by merging business practices in public sector affairs. The design of the nursing home planning system has broadband implications.

It intersects legal, theoretical, and practical consequences, enforcing web-based systems as the suitable basis to weave intersectoral communication and fragmented systems to achieve NPM principles on effective public service delivery (Hood, 1991, p. 5), The final result provides a one-stop service or end-to-end nursing home planning system application to reduce dispersed means of intersectoral interaction found in the current planning system. Reflecting on Hevner and Peffer, the iteration on the design aims to maximize user experience (UX). Thus, the design was supplied by toggletip to improve ease of use.

Analyzing the software market, it is believed that low-code is not necessarily the best and potentially requires thorough synergy with other systems to complement rollout and long-term stability. Overall, this proposed design offers a blueprint to digitalize the nursing home planning system that is modular, integrated, seamless and data-driven. This research proposes alternatives that support flexibility. Regardless, for crucial parts found in the system or access permission, full custom coding is required to enhance security and trust among the stakeholders. A hybrid approach that uses both methods helps accelerate the rollout and keeps things sustainable and profitable in the long run.

This research ensures that the system complies with the EU context by adhering to relevant institutional inertia, mainly OZG. User centricity is largely the main approach in designing nursing home planning systems, as the public sector lacks technocratic ability. Although the developed artifact remains conceptual and was not implemented in a real-world setting, this research addresses this chasm by embracing structured and stakeholder-informed design for future implementation. However, this

research is limited by the number of respondents verifying and validating the evaluation. Nonetheless, the countermeasure is adopting the most practical sense-making concept and proposal to produce a design with high usability.

This research is only limited to proposing design and not including the implementation plan and management. Future researchers are recommended to explore path dependence in Germany's digitalization and test the system execution to experiment with the feasibility of the proposed design. As this research is constrained to the available use case, it is limited in capturing official regulation and external factors contributing to the system. Future research can address this shortcoming to create robust design science research. The semantic understanding is not addressed and becomes another critical part for future reference. Moreover, it opens to constructive view on the possibility of this system diffuse to citizens' cooperation or co-creation approach that similarly reflected by participatory e-budgeting

References

Andersson, E. (2014). Engagement in health: roles for the public and patients. Smart governance for health and well-being: the evidence, 34

Asgarkhani, M. (2007). Public Management Review DIGITAL GOVERNMENT AND ITS EFFECTIVENESS IN PUBLIC MANAGEMENT REFORM A local government perspective. Public Management Review, 7(3), 465–487. https://doi.org/10.1080/14719030500181227

Ashar, M., Haliah, H., & Nirwana, N. (2025). ERP in Public Sector Reform: A Systematic Literature Review of Technological, Organizational, and Institutional Factors. Journal Research of Social Science, Economics, and Management, 4(10), 1420–1439. https://doi.org/10.59141/jrssem.v4i10.810

Audretschh·, D., Lehmann, E., & Vismara, A. S. (n.d.). Globalization and Public Policy A European Perspective.

Becker, J., Algermissen, L., & Niehaves, B. (n.d.). A procedure model for process oriented e-government projects. https://doi.org/10.1108/14637150610643760

Bing Wu, Lawless, D., Bisbal, J., Grimson, J., Wade, V., O'Sullivan, D., & Richardson, R. (n.d.). Legacy systems migration-a method and its tool-kit framework. Proceedings of Joint 4th International Computer Science Conference and 4th Asia Pacific Software Engineering Conference, 312–320. https://doi.org/10.1109/apsec.1997.640188

Buko, J. (2018). Premises and conditions for the implementation of government-certified e-communication on the example of selected EU countries. European Journal of Service Management, 28, 49–54.

Cai, L., & Zhu, Y. (2015). The Challenges of Data Quality and Data Quality Assessment in the Big Data Era. Data Science Journal, 14(0), 2. https://doi.org/10.5334/dsj-2015-002

Cao, G., Duan, Y., & Cadden, T. (2018). The link between information processing capability and competitive advantage mediated through decision-making effectiveness. https://doi.org/10.1016/j.ijinfomgt.2018.10.003

Carroll, J. M., & Ganoe, C. H. (2008). Supporting community with location-sensitive mobile applications. Handbook of Research on Urban Informatics: The Practice and Promise of the Real-Time City, 339–352. https://doi.org/10.4018/978-1-60566-152-0.CH023

Cavallo, A., Sanasi, S., Ghezzi, A., & Rangone, A. (n.d.). Competitive intelligence and strategy formulation: connecting the dots. https://doi.org/10.1108/CR-01-2020-0009

Chou, D. C., Bindu Tripuramallu, H., & Chou, A. Y. (2005). BI and ERP integration. Information Management & Computer Security, 13(5), 340–349. https://doi.org/10.1108/09685220510627241

Chuanfan, L. (2010). Research on Role-Based Access Control Policy of E-government. 2010 International Conference on E-Business and E-Government, 714–716. https://doi.org/10.1109/ICEE.2010.186

De Almeida Mello, J., Wellens, N. I., Hermans, K., De Stampa, M., Cerase, V., Vereker, N., Jónsson, P. V., Finne-Soveri, H., Meehan, B., & Declercq, A. (2023). The Implementation of Integrated Health Information Systems – Research Studies from 7 Countries Involving the InterRAI Assessment System. International Journal of Integrated Care, 23, 8. pg (De Almeida Mello et al., 2023)

De Montis, A. (2016). European Planning Studies Measuring the performance of planning: the conformance of Italian landscape planning practices with the European Landscape Convention Measuring the performance of planning: the conformance of Italian landscape planning practices with the European Landscape Convention. https://doi.org/10.1080/09654313.2016.1178215

Demography Report. (n.d.).

Devika Rajhamundry. (2024). Business Intelligence as a Strategic Planning Tool in Healthcare Organizations: A Systems Integration Approach. International Journal of Scientific Research in Computer Science, Engineering and Information Technology, 10(6), 1965–1972. https://doi.org/10.32628/cseit241061234

Ekeland, A. G., Line, ;, & Linstad, H. (n.d.). Elaborating Models of eHealth Governance: Qualitative Systematic Review. https://doi.org/10.2196/17214

Elbashir, M. Z., Sutton, S. G., Arnold, V., Collier, P. A., Janvrin, D., Sivabalan, P., & Widener, S. (2022). Leveraging business intelligence systems to enhance management control and business process performance in the public sector. Meditari Accountancy Research, 30(4), 914–940. https://doi.org/10.1108/MEDAR-04-2021-1287

Fahr, M. C., Förster, · Maximilian, Moestue, L., Brasse, J., Klier, J., & Klier, · Mathias. (123 C.E.). Improving the use of public e-services through explainability. Journal of Business Economics, 95, 553–586. https://doi.org/10.1007/s11573-024-01212-9

Ferraiolo, D. F., Sandhu, R., Gavrila, S., Kuhn, D. R., & Chandramouli, R. (2001). Proposed NIST standard for role-based access control. *ACM Transactions on Information and System Security*, *4*(3), 224–274. https://doi.org/10.1145/501978.501980

Foth, M., Brynskov, M., & Ojala, T. (2015a). Citizen's right to the digital city urban interfaces, activism, and placemaking. Citizen's Right to the Digital City: Urban Interfaces, Activism, and Placemaking, 1–259. https://doi.org/10.1007/978-981-287-919-6

Ghezzi, R., Kolehmainen, T., Setälä, M., & Mikkonen, T. (2024). Enterprise Architecture as an Enabler for a Government Business Ecosystem: Experiences from Finland. In R. Chbeir, D. Benslimane, M. Zervakis, Y. Manolopoulos, N. T. Ngyuen, & J. Tekli (Eds.), Management of Digital EcoSystems (Vol. 2022, pp. 219–233). Springer Nature Switzerland. https://doi.org/10.1007/978-3-031-51643-6_16

Gonin, A., & Neumann, O. (2024). Does Introducing FAQs Boost Self-Service Government?—A Study of Local Government Websites. International Journal of Public Administration. https://doi.org/10.1080/01900692.2024.2405735/ASSET/5EED66B3-76B1-4D2C-BE6B-B078ADC8A3FB/ASSETS/GRAPHIC/LPAD_A_2405735_F0002_B.GIF

Gräfe, P., Marienfeldt, J., Wehmeier, L. M., & Kuhlmann, S. (n.d.). Changing tasks and changing public servants? The digitalisation and automation of public administrative work. https://doi.org/10.1177/13837605241289773

Güner, E. O., Han, S., & Juell-Skielse, G. (2022). Enhancing routine capability through robotic process automation in the public sector: a case survey. In Progress in IS (pp. 169–188). https://doi.org/10.1007/978-3-030-92644-1_9

Harris, S., Gibbons, J., Davies, J., Tsui, A., & Crichton, C. (2008). Semantic technologies in electronic government. Proceedings of the 2nd International

Conference on Theory and Practice of Electronic Governance, 45–51. https://doi.org/10.1145/1509096.1509108

Health expenditures in the EU 2022—German Federal Statistical Office. (n.d.). Retrieved 31 July 2025, from https://www.destatis.de/Europa/EN/Topic/Population-Labour-Social-Issues/Health/HealthExpenditure.html

HeimG - Home law. (n.d.). Retrieved 1 June 2025, from https://www.gesetze-im-internet.de/heimg/BJNR018730974.html

Hevner, A. R., March, S. T., Park, J., & Ram, S. (n.d.). *Design Science in Information Systems Research*.

Hindarto, D., & Wedha, B. Y. (2023). Maximizing ERP Benefits with Enterprise Architecture: A Holistic Approach. Journal of Computer Networks, Architecture and High Performance Computing, 5(2), 703–713. https://doi.org/10.47709/cnahpc.v5i2.2790

Home laws of the federal states | BIVA Nurgeschutzbund. (2014, February 24). https://www.biva.de/deutsches-pflegesystem/gesetze/laender-heimgesetze/

Hood, C. (1991). A PUBLIC MANAGEMENT FOR ALL SEASONS? Public Administration (London), 69(1), 3–19. https://doi.org/10.1111/j.1467-9299.1991.tb00779.x

Hou, C. K., & Papamichail, K. N. (2010). The impact of integrating enterprise resource planning systems with business intelligence systems on decision-making performance: An empirical study of the semiconductor industry. International Journal of Technology, Policy and Management, 10(3), 201. https://doi.org/10.1504/ijtpm.2010.034510

Hustad, E., & Stensholt, J. (2023). Customizing ERP-systems: A framework to support the decision-making process. Procedia Computer Science, 219, 789–796. https://doi.org/10.1016/j.procs.2023.01.352

Immonen, A., Palviainen, M., & Ovaska, E. (2014). Requirements of an open data based business ecosystem. IEEE Access, 2, 88–103. https://doi.org/10.1109/ACCESS.2014.2302872

Irani, Z., Abril, R. M., Weerakkody, V., Omar, A., & Sivarajah, U. (2023). The impact of legacy systems on digital transformation in European public administration:

Lesson learned from a multi case analysis. Government Information Quarterly, 40(1), 101784. https://doi.org/10.1016/j.giq.2022.101784

Jaradat, Z., AL-Hawamleh, A., & Hamdan, A. (n.d.). Examining the integration of ERP and BI in the industrial sector and its impact on decision-making processes in KSA. https://doi.org/10.1108/DPRG-04-2024-0077

Ji, Y., Zhang, P., Qiu, C., Li, P., & Wang, D. (2022). How TOGAF® can address government interoperability problems A case study of the government of Canada.

Ji, Y.-A., & Kim, H.-S. (2022). Scoping Review of the Literature on Smart Healthcare for Older Adults. Yonsei Medical Journal, 63(Suppl), S14–S21.

John, Y. (2024). Does Enterprise Resource Planning (ERP) make a difference in public sector organisations' data management? Lessons from the Ministry of Lands, Housing and Human

Jordan, S., Zabukovšek, S. S., & Klančnik, I. Š. (2022). Document Management System – A Way to Digital Transformation. Naše Gospodarstvo/Our Economy, 68(2), 43–54. https://doi.org/10.2478/ngoe-2022-0010

Kashyap, V., Morales, A., & Hongsermeier, T. (2006). On implementing clinical decision support: achieving scalability and maintainability by combining business rules and ontologies.

Kelemen, R. (2014). ERP systems in public sector. 2014 37th International Convention on Information and Communication Technology, Electronics and Microelectronics (MIPRO), 1537–1543. https://doi.org/10.1109/MIPRO.2014.6859810

Kencana, A. S., Fajrillah, A. A. N., & Hanafi, R. (2022). DESIGN OF ENTERPRISE ARCHITECTURE IN E-GOVERNMENT DEVELOPMENT USING TOGAF ADM 9.2 FRAMEWORK IN ELECTRICITY SECTOR OF DEPARTMENT ENERGY AND MINERAL RESOURCE OF WEST JAVA PROVINCE. JIPI (Jurnal Ilmiah Penelitian Dan Pembelajaran Informatika), 7(4), 1068–1079. https://doi.org/10.29100/jipi.v7i4.3203

Kern, C. J., Noeltner, M., & Kroenung, J. (n.d.). The Case of Digital Ethics in IS Research - A Literature Review. International Conference on Information Systems, ICIS 2022: "Digitization for the Next Generation," Copenhagen.

Khumalo, S., & Mearns, M. (2019). SharePoint as enabler for collaboration and efficient project knowledge sharing. SA Journal of Information Management, 21(1). https://doi.org/10.4102/sajim.v21i1.1044

Kim, J. Y., Boag, W., Gulamali, F., Hasan, A., David, H., Hogg, J., Lifson, M., Mulligan, D., Patel, M., Raji, I. D., Sehgal, A., Shaw, K., Tobey, D., Valladares, A., Vidal, D., Balu, S., Sendak, M., & Raji, D. (n.d.). Organizational Governance of Emerging Technologies: AI Adoption in Healthcare. 23. https://doi.org/10.1145/3593013.3594089

Ko, M., Wagner, L., & Spetz, J. (2018a). Nursing Home implementation of health information Technology: Review of the literature finds inadequate investment in preparation, infrastructure, and training. INQUIRY the Journal of Health Care Organization Provision and Financing, 55. https://doi.org/10.1177/0046958018778902

Ko, M., Wagner, L., & Spetz, J. (2018b). Nursing home implementation of health information technology: Review of the literature finds inadequate investment in preparation, infrastructure, and training. Inquiry (United States), 55, 1–10. https://doi.org/10.1177/0046958018778902/ASSET/7167893D-9FC3-442F-8BDB-866CB30B24C9/ASSETS/IMAGES/LARGE/10.1177_0046958018778902-FIG1.JPG

Kolawole Joseph Ajiboye. (2021). The role of Low-Code/No-Code platforms in accelerating digital transformation in regulated industries. International Journal of Science and Research Archive, 4(1), 262–279. https://doi.org/10.30574/ijsra.2021.4.1.0159

Krug, S. (2014). Don’t make me think, revisited: a common sense approach to web usability (3rd edition). New Riders.

Kuhn, K. (2011). Open government data and public transportation. Journal of Public Transportation, 14(1), 83–97. https://doi.org/10.5038/2375-0901.14.1.5

Kumar, V., Maheshwari, B., Kumar, & U., Kumary, V., Maheshwariy, B., & Kumary, U. (2002). International Journal of Production Research Enterprise resource planning systems adoption process: A survey of Canadian organizations Enterprise resource planning systems adoption process: a survey of Canadian organizations. 40(3). https://doi.org/10.1080/00207540110092414

Kwahk, K. Y., & Ahn, H. (2010a). Moderating effects of localization differences on ERP use: A socio-technical systems perspective. Computers in Human Behavior, 26(2), 186–198. https://doi.org/10.1016/J.CHB.2009.10.006

Kwahk, K. Y., & Ahn, H. (2010b). Moderating effects of localization differences on ERP use: A socio-technical systems perspective. Computers in Human Behavior, 26(2), 186–198. https://doi.org/10.1016/J.CHB.2009.10.006

Lagzi, M. D., Farkhondeh, F., Mahdiraji, H. A., & Sakka, G. (n.d.). Exploring data-driven decision-making practices: a comprehensive review with bibliometric insights and future directions. https://doi.org/10.1108/EMJB-11-2024-0307

Lapão, L. V., da Silva, M. M., & Gregório, J. (2017). Implementing an online pharmaceutical service using design science research. *BMC Medical Informatics and Decision Making*, *17*(1), 31. https://doi.org/10.1186/s12911-017-0428-2

Li, J., Wu, T., Liu, B., & Zhou, M. (2024). Can digital transformation enhance corporate ESG performance? The moderating role of dual environmental regulations. *Finance Research Letters*, *62*, Article 105241. https://doi.org/10.1016/j.frl.2024.105241

Li, Y., Fan, Y., Nie, L., & Li, Y. (2025). Making governance agile: Exploring the role of artificial intelligence in China's local governance. Public Policy and Administration, 40(2), 276–301. https://doi.org/10.1177/09520767231188229

Longo, J. (2024). The Transformative Potential of Artificial Intelligence for Public Sector Reform. Canadian Public Administration. https://doi.org/10.1111/CAPA.12587

Long-term care projection: 1.8 million more people in need of long-term care expected until 2055. (n.d.). Federal Statistical Office. Retrieved 31 July 2025, from https://www.destatis.de/EN/Press/2023/03/PE23_124_12.html

Mergel, I. (2013). Social media adoption and resulting tactics in the U.S. federal government. Government Information Quarterly, 30(2), 123–130. https://doi.org/10.1016/J.GIQ.2012.12.004

Meutia, I. F., Yulianti, D., & Sujadmiko, M. B. (2024). Driving Innovative Healthcare Reform through Meta-governance Strategies. Health Behav Policy Rev.TM, 11(4), 1613–1623. https://doi.org/10.14485/HBPR.11.4.1

Michaeli, D. T., Hasan, ;, Yagmur, B., Michaeli, ; Thomas, & Michaeli, T. (n.d.). Germany's healthcare system: Funding, resource allocation, provider payment,

efficiency, and access. Retrieved June 4, 2025, from https://ssrn.com/abstract=4072939

Mohammed, A., & Saleh, B. M. (2017). Centralized Database: A Prerequisite for Security and Sustainable Development in Nigeria. International Journal of Innovative Research in Computer Science & Technology, 5(1), 209–213. https://doi.org/10.21276/ijircst.2017.5.1.7

Moussas, K., Hafiane, J., & Achaba, A. (2024). Business intelligence and its pivotal role in organizational performance: An exhaustive literature review. Journal of Autonomous Intelligence, 7(4). https://doi.org/10.32629/jai.v7i4.1286

Murthy, K. K. K. (2021). Transforming Legacy Systems: Strategies For Successful ERP Implementations In Large Organizations. 9(6).

Nfuka, E. N., & Rusu, L. (n.d.). The effect of critical success factors on IT governance performance. https://doi.org/10.1108/02635571111182773

Ng, C. S.-P. (2013). A Case Study on the Impact of Customization, Fitness, and Operational Characteristics on Enterprise-Wide System Success, User Satisfaction, and System Use. Journal of Global Information Management, 21(1), 19–41. https://doi.org/10.4018/jgim.2013010102

Nofal, M. I., & Yusof, Z. M. (2013). Integration of Business Intelligence and Enterprise Resource Planning within Organizations. Procedia Technology, 11, 658–665. https://doi.org/10.1016/j.protcy.2013.12.242

Nursing Home Implementation of Health Information Technology: Review of the Literature Finds Inadequate Investment in Preparation, Infrastructure, and Training. (n.d.). Retrieved July 12, 2025, from https://journals.sagepub.com/doi/epdf/10.1177/0046958018778902?src=getftr&utm_source=mendeley&getft_integrator=mendeley

Ovalle, C., Lezcano, G. C., & Luna, R. F. (2024). Design of an Enterprise Architecture Model Applying TOGAF in the Sales Management of Fast Food SMEs. International Journal of Engineering Trends and Technology, 72(3), 331–348. https://doi.org/10.14445/22315381/IJETT-V72I3P129

Pang, M. S., Lee, G., & Delone, W. H. (2014). In public sector organisations: A public-value management perspective. Journal of Information Technology, 29(3), 187–205. https://doi.org/10.1057/JIT.2014.2;

Parrado, S. (n.d.). Failed policies but institutional innovation through “layering” and “diffusion” in Spanish central administration. https://doi.org/10.1108/09513550810855672

Parthe, R. M. (2023). Comparative Analysis of Data Visualization Tools: Power BI and Tableau. INTERANTIONAL JOURNAL OF SCIENTIFIC RESEARCH IN ENGINEERING AND MANAGEMENT, 07(10), 1–11. https://doi.org/10.55041/ijsrem26272

Peffers, K., Tuunanen, T., Rothenberger, M. A., & Chatterjee, S. (2007). A Design Science Research Methodology for Information Systems Research. *Journal of Management Information Systems*, *24*(3), 45–77. https://doi.org/10.2753/MIS0742-1222240302

Petter, S., Delone, W., & Mclean, E. (2008). European Journal of Information Systems Measuring information systems success: models, dimensions, measures, and interrelationships Measuring information systems success: models, dimensions, measures, and interrelationships. European Journal of Information Systems, 17, 236–263. https://doi.org/10.1057/ejis.2008.15

Pflegeversicherung. (n.d.). BMG. Retrieved 31 May 2025, from https://www.bundesgesundheitsministerium.de/themen/pflege/online-ratgeber-pflege/die-pflegeversicherung.html

Population by age group and variant until 2070. (n.d.). Federal Statistical Office. Retrieved 31 July 2025, from https://www.destatis.de/EN/Themes/Society-Environment/Population/Population-Projection/Tables/variant-1-2-3-agegroups.html

Public Administration in Germany. (2021). [Book]. In S. Kuhlmann, I. Proeller, D. Schimanke, & J. Ziekow (Eds.), Governance and Public Management.

Purchase, V., Parry, G., Valerdi, R., Nightingale, D., & Mills, J. (2011). Journal of Enterprise Transformation Enterprise Transformation: Why Are We Interested, What Is It, and What Are the Challenges? ENTERPRISE TRANSFORMATION: WHY ARE WE INTERESTED, WHAT IS IT, AND WHAT ARE THE CHALLENGES? https://doi.org/10.1080/19488289.2010.549289

Raso, J., Adelmant, V., Bhuta, N., Johns, F., Sheffi, N., Streinz, T., & Sullivan, G. (2023). Smooth Operators, Predictable Glitches: The Interface Governance of

Benefits and Borders. Canadian Journal of Law and Society / Revue Canadienne Droit et Société, 38(2), 158–179. https://doi.org/10.1017/cls.2023.20

Sandhu, R., Ferraiolo, D., & Kuhn, R. (2000). The NIST model for role-based access control: Towards a unified standard. *Proceedings of the Fifth ACM Workshop on Role-Based Access Control*, 47–63. https://doi.org/10.1145/344287.344301

Sandhu, R. S., Coyne, E. J., Feinstein, H. L., & Youman, C. E. (1996). Role-based access control models. *Computer*, *29*(2), 38–47. https://doi.org/10.1109/2.485845

Sandoval-Almazan, R., & Gil-Garcia, J. R. (2011). Are government internet portals evolving towards more interaction, participation, and collaboration? Revisiting the rhetoric of e-government among municipalities. https://doi.org/10.1016/j.giq.2011.09.004

Sang, S., Jeong-Dong, L., & Lee, J. (2009). Adoption of E-Government Services: The Case of Electronic Approval System. International Journal of E-Adoption, 1(2), 1-22. https://doi.org/10.4018/jea.2009040101

Schmidt-Kraepelin, C., Birgit, A. E., Ae, J., & Gaebel, W. (n.d.). Prevention of rehospitalization in schizophrenia: results of an integrated care project in Germany. https://doi.org/10.1007/s00406-009-0056-7

Schmitt, T., Haarmann, A., & Shaikh, M. (n.d.). Strengthening health system governance in Germany: looking back, planning ahead. https://doi.org/10.1017/S1744133122000123

Scholl, H. J. (2012). Five trends that matter: Challenges to 21st century electronic government. Information Polity, 17(3–4), 317–327. https://doi.org/10.3233/IP-2012-0280

Seidel, S., Recker, J., & vom Brocke, J. (2013). Sensemaking and Sustainable Practicing: Functional Affordances of Information Systems in Green Transformations. MIS Quarterly, 37(4), 1275–1299. https://doi.org/10.25300/MISQ/2013/37.4.13

SGB V Statutory Health Insurance—SGB V Social Code. (n.d.). Retrieved 31 May 2025, from https://www.sozialgesetzbuch-sgb.de/sgbv/1.html

SGB XI Soziale Pflegeversicherung—SGB XI Sozialgesetzbuch. (n.d.). Retrieved 31 May 2025, from https://www.sozialgesetzbuch-sgb.de/sgbxi/1.html

Shang, S., & Seddon, P. B. (2002). Assessing and managing the benefits of enterprise systems: The business manager's perspective. Information Systems Journal, 12(4), 271–299. https://doi.org/10.1046/j.1365-2575.2002.00132.x

Shi, Z., Dong, J., & Gan, Y. (2025). Democratizing Digital Transformation: A Multisector Study of Low-Code Adoption Patterns, Limitations, and Emerging Paradigms. Applied Sciences, 15(12), 6481. https://doi.org/10.3390/app15126481

Simonsson, M., Johnson, P., & Ekstedt, M. (2010a). Information Systems Management The Effect of IT Governance Maturity on IT Governance Performance. https://doi.org/10.1080/10580530903455106

Simonsson, M., Johnson, P., & Ekstedt, M. (2010b). Information Systems Management The Effect of IT Governance Maturity on IT Governance Performance. https://doi.org/10.1080/10580530903455106

Sinard, J. H., & Gershkovich, P. (2012). Custom software development for use in a clinical laboratory. Journal of Pathology Informatics, 3(1), 44. https://doi.org/10.4103/2153-3539.104906

Singu, S. K. (2022). ETL Process Automation: Tools and Techniques.

Smart governance for health and well-being: the evidence. (n.d.-a).

Smart governance for health and well-being: the evidence. (n.d.-b).

Sommer, S., & Proff, H. (2023). Digital Maturity in the German Traditional Industries — Status Quo, Profit Impact, and Paths of Acceleration. International Journal of Innovation and Technology Management, 20(4), Article 2350019. https://doi.org/10.1142/S0219877023500190

Sreenivasa Rao Sola. (2021). ERP Migration in Digital Transformation: Best Practices and Overcoming Integration Challenges. https://doi.org/10.5281/ZENODO.15259072

Statista. (n.d.). Business Intelligence Software – Germany. In Statista Outlook: Enterprise software. Retrieved August 1, 2025, from https://www.statista.com/outlook/tmo/software/enterprisesoftware/businessintelligencesoftware/germany#revenue

Sternad Zabukovšek, S., Tominc, P., Deželak, Z., Nalbandyan, G., & Bobek, S. (2022). Acceptance of GIS within ERP System: Research Study in Higher Education.

ISPRS International Journal of Geo-Information, 11(2), 83. https://doi.org/10.3390/ijgi11020083

Stiel, S., Brütt, A. L., Stahmeyer, J. T., Bockelmann, A. W. E., Schleef, T., Völkel, A., & Hoffmann, F. (2023). Implementation, barriers, and recommendations for further development of advance care planning for the last phase of life in nursing homes in Germany (Gut-Leben): protocol for a mixed-methods study. BMC Palliative Care, 22(1), 1–9. https://doi.org/10.1186/S12904-023-01147-Y/FIGURES/2

Stoker, G. (1998). Governance as theory: five propositions. International social science journal, 50(155), 17-28.

Sundberg, L., Floren, H., & Sundberg, H. (2023). Enterprise architecture adoption in government: A public value perspective. Proceedings of the 16th International Conference on Theory and Practice of Electronic Governance, 254–262. https://doi.org/10.1145/3614321.3614356

Szukits, Á., Móricz, P., Móricz petermoricz, P., Szukits, Á., & Móricz, P. (2023). Towards data-driven decision making: the role of analytical culture and centralization efforts. Review of Managerial Science 2023 18:10, 18(10), 2849–2887. https://doi.org/10.1007/S11846-023-00694-1

Sá nchez, A., Baladró, C., Aguiar, J., Carro, B., Goix, L.-W., rgen Sienel, J., Ordá, I., Trapero, R., & Bascuñ ana, A. (n.d.). 1 User-centric Service Creation and Execution. https://doi.org/10.7551/mitpress/7959.001.0001

The Open Group. (2022). TOGAF® Standard, 10th Edition: Introduction and Core Concepts. https://pubs.opengroup.org/togaf-standard/core/

The Scalability-Efficiency/Maintainability-Portability Trade-Off in Simulation Software Engineering: Examples and a Preliminary Systematic Literature review. (2016, November 1). IEEE Conference Publication | IEEE Xplore. https://ieeexplore.ieee.org/document/7839468

Triyanto, T., & Supriyanto, A. (2024). Enterprise Resource Planning (ERP) Design Using TOGAF ADM and ACMM (Case Study: PT XYZ). Information Technology International Journal, 2(1). https://doi.org/10.33005/itij.v2i1.16

Van Der Waal, S., W Ę Cel, K., Ermilov, I., Janev, V., Milošević, U., & Wainwright, M. (2014). Lifting open data portals to the data web. Lecture Notes in Computer Science (Including Subseries Lecture Notes in Artificial Intelligence and Lecture

Notes in Bioinformatics), 8661, 175–195. https://doi.org/10.1007/978-3-319-09846-3_9

Van Loenen, B. (2018). The Development of Open Data in The Netherlands. 215–232. https://doi.org/10.1007/978-94-6265-261-3_11

Vanneste, D., & Declercq, A. (2014). The development of BelRAI, a web application for sharing assessment data on frail older people in home care, nursing homes, and hospitals. In Advances in healthcare information systems and administration book series (pp. 202–226). https://doi.org/10.4018/978-1-4666-6138-7.ch010

Veit, D. J., Parasie, N. P., & Schwarz Chair, D. (2009). Can National E-Government Standards find Acceptance?

View of Legal and Organizational Economic Aspects of the Functioning of the Main Models of Health-Care Systems. (n.d.). Retrieved June 4, 2025, from https://www.asiapharmaceutics.info/index.php/ajp/article/view/2631/996

Wagner, W., & Antonucci, Y. L. (2009). The ImaginePA Project: The First Large-Scale, Public Sector ERP Implementation. Information Systems Management, 26(3), 275–284. https://doi.org/10.1080/10580530903017401

Weggeman, M., & Cauffman, L. (2024). Using design science research to develop and validate the application of client-oriented psychological approaches illustrated by the design of a solution-focused approach. Methods in Psychology, 11. https://doi.org/10.1016/j.metip.2024.100151

Williamson, B. (2014). Knowing public services: Cross-sector intermediaries and algorithmic governance in public sector reform. Public Policy and Administration, 29(4), 292–312. https://doi.org/10.1177/0952076714529139/FORMAT/EPUB

Willke, H. (2007). Smart governance: governing the global knowledge society. Campus Verlag.

Zheng, J., & Khalid, H. (2022). The Adoption of Enterprise Resource Planning and Business Intelligence Systems in Small and Medium Enterprises: A Conceptual Framework. Mathematical Problems in Engineering, 2022. https://doi.org/10.1155/2022/1829347

Appendix

Appendix 1: Interview Transcript

The link below shows the 20-page interview transcript.

A link to the interview transcript

Appendix 2: Questions Guidelines

- Is the proposed architecture technically feasible and scalable for the public sector?
- Which components seem most practical, and which appear overly complex?
- Do the diagrams clearly and professionally represent the proposed solution?
- What aspects of the design could be improved?
- Does the design comply with relevant regulations, or are there potential violations?
- Which regulations should we specifically consider?
- Are there any integration challenges we haven't already addressed?
- How well does the role-based access control align with the use case and overall design?
- What is your perspective on the data migration approach (tentative)?
- Are there any essential tools or technologies you expected to see but are missing?
- Does the system design accurately reflect all current process flows?

Declaration of Authorship

I hereby declare that, to the best of my knowledge and belief, this Seminar Thesis titled **“ERP-inspired Nursing Home Planning System”** is my own work. I confirm that each significant contribution to and quotation in this thesis that originates from the work or works of others is indicated by proper use of citation and references.

Münster, 02 August 2025

Abishek Sivakumar

Declaration of Authorship

I hereby declare that, to the best of my knowledge and belief, this Seminar Thesis titled **“ERP-inspired Nursing Home Planning System”** is my own work. I confirm that each significant contribution to and quotation in this thesis that originates from the work or works of others is indicated by proper use of citation and references.

Münster, 02 August 2025

Aizhan Kanybekova

Declaration of Authorship

I hereby declare that, to the best of my knowledge and belief, this Master/Bachelor/Seminar Thesis titled **"ERP-inspired Nursing Home Planning System"** is my own work. I confirm that each significant contribution to and quotation in this thesis that originates from the work or works of others is indicated by proper use of citation and references.

Münster, 02 August 2025

Chalisa Ivanka Salsabila

Declaration of Authorship

I hereby declare that, to the best of my knowledge and belief, this Seminar Thesis titled **“ERP-inspired Nursing Home Planning System”** is my own work. I confirm that each significant contribution to and quotation in this thesis that originates from the work or works of others is indicated by proper use of citation and references.

Münster, 02 August 2025

Lorenzo Lazarus

Consent Form

for the use of plagiarism detection software to check my thesis

Name: Sivakumar **Given Name:** Abishek
Student number:563736 **Course of Study:** PIONEER
Address: Schlossplatz 2, 48149 Münster
Title of the thesis: "ERP-inspired Nursing Home Planning System"

What is plagiarism? Plagiarism is defined as submitting someone else's work or ideas as your own without a complete indication of the source. It is hereby irrelevant whether the work of others is copied word by word without acknowledgment of the source, text structures (e.g. line of argumentation or outline) are borrowed or texts are translated from a foreign language.

Use of plagiarism detection software. The examination office uses plagiarism software to check each submitted bachelor and master thesis for plagiarism. For that purpose the thesis is electronically forwarded to a software service provider where the software checks for potential matches between the submitted work and work from other sources. For future comparisons with other theses, your thesis will be permanently stored in a database. Only the School of Business and Economics of the University of Münster is allowed to access your stored thesis. The student agrees that his or her thesis may be stored and reproduced only for the purpose of plagiarism assessment. The first examiner of the thesis will be advised on the outcome of the plagiarism assessment.

Sanctions. Each case of plagiarism constitutes an attempt to deceive in terms of the examination regulations and will lead to the thesis being graded as "failed". This will be communicated to the examination office where your case will be documented. In the event of a serious case of deception the examinee can be generally excluded from any further examination. This can lead to the exmatriculation of the student. Even after completion of the examination procedure and graduation from university, plagiarism can result in a withdrawal of the awarded academic degree.

I confirm that I have read and understood the information in this document. I agree to the outlined procedure for plagiarism assessment and potential sanctioning.

Münster, 02 August 2025

Abishek Sivakumar

Consent Form

for the use of plagiarism detection software to check my thesis

Name: Kanybekova **Given Name:** Aizhan
Student number: 563723 **Course of Study:** PIONEER
Address: Schlossplatz 2, 48149 Münster
Title of the thesis: "ERP-inspired Nursing Home Planning System"

What is plagiarism? Plagiarism is defined as submitting someone else's work or ideas as your own without a complete indication of the source. It is hereby irrelevant whether the work of others is copied word by word without acknowledgment of the source, text structures (e.g. line of argumentation or outline) are borrowed or texts are translated from a foreign language.

Use of plagiarism detection software. The examination office uses plagiarism software to check each submitted bachelor and master thesis for plagiarism. For that purpose the thesis is electronically forwarded to a software service provider where the software checks for potential matches between the submitted work and work from other sources. For future comparisons with other theses, your thesis will be permanently stored in a database. Only the School of Business and Economics of the University of Münster is allowed to access your stored thesis. The student agrees that his or her thesis may be stored and reproduced only for the purpose of plagiarism assessment. The first examiner of the thesis will be advised on the outcome of the plagiarism assessment.

Sanctions. Each case of plagiarism constitutes an attempt to deceive in terms of the examination regulations and will lead to the thesis being graded as "failed". This will be communicated to the examination office where your case will be documented. In the event of a serious case of deception the examinee can be generally excluded from any further examination. This can lead to the exmatriculation of the student. Even after completion of the examination procedure and graduation from university, plagiarism can result in a withdrawal of the awarded academic degree.

I confirm that I have read and understood the information in this document. I agree to the outlined procedure for plagiarism assessment and potential sanctioning.

Münster, 02 August 2025

Aizhan Kanybekova

Consent Form

for the use of plagiarism detection software to check my thesis

Name: Salsabila **Given Name:** Chalisa Ivanka
Student number: 563734 **Course of Study:** PIONEER
Address: Stenfurter Strasse 79, 48149 Münster
Title of the thesis: "ERP-inspired Nursing Home Planning System"

What is plagiarism? Plagiarism is defined as submitting someone else's work or ideas as your own without a complete indication of the source. It is hereby irrelevant whether the work of others is copied word by word without acknowledgment of the source, text structures (e.g. line of argumentation or outline) are borrowed or texts are translated from a foreign language.

Use of plagiarism detection software. The examination office uses plagiarism software to check each submitted bachelor and master thesis for plagiarism. For that purpose the thesis is electronically forwarded to a software service provider where the software checks for potential matches between the submitted work and work from other sources. For future comparisons with other theses, your thesis will be permanently stored in a database. Only the School of Business and Economics of the University of Münster is allowed to access your stored thesis. The student agrees that his or her thesis may be stored and reproduced only for the purpose of plagiarism assessment. The first examiner of the thesis will be advised on the outcome of the plagiarism assessment.

Sanctions. Each case of plagiarism constitutes an attempt to deceive in terms of the examination regulations and will lead to the thesis being graded as "failed". This will be communicated to the examination office where your case will be documented. In the event of a serious case of deception the examinee can be generally excluded from any further examination. This can lead to the exmatriculation of the student. Even after completion of the examination procedure and graduation from university, plagiarism can result in a withdrawal of the awarded academic degree.

I confirm that I have read and understood the information in this document. I agree to the outlined procedure for plagiarism assessment and potential sanctioning.

Münster, 02 August 2025

Chalisa Ivanka Salsabila

Consent Form

for the use of plagiarism detection software to check my thesis

Name: Lazarus **Given Name:** Lorenzo
Student number: 563725 **Course of Study:** PIONEER
Address: Schlossplatz 2, 48149 Münster
Title of the thesis: "ERP-inspired Nursing Home Planning System"

What is plagiarism? Plagiarism is defined as submitting someone else's work or ideas as your own without a complete indication of the source. It is hereby irrelevant whether the work of others is copied word by word without acknowledgment of the source, text structures (e.g. line of argumentation or outline) are borrowed or texts are translated from a foreign language.

Use of plagiarism detection software. The examination office uses plagiarism software to check each submitted bachelor and master thesis for plagiarism. For that purpose the thesis is electronically forwarded to a software service provider where the software checks for potential matches between the submitted work and work from other sources. For future comparisons with other theses, your thesis will be permanently stored in a database. Only the School of Business and Economics of the University of Münster is allowed to access your stored thesis. The student agrees that his or her thesis may be stored and reproduced only for the purpose of plagiarism assessment. The first examiner of the thesis will be advised on the outcome of the plagiarism assessment.

Sanctions. Each case of plagiarism constitutes an attempt to deceive in terms of the examination regulations and will lead to the thesis being graded as "failed". This will be communicated to the examination office where your case will be documented. In the event of a serious case of deception the examinee can be generally excluded from any further examination. This can lead to the exmatriculation of the student. Even after completion of the examination procedure and graduation from university, plagiarism can result in a withdrawal of the awarded academic degree.

I confirm that I have read and understood the information in this document. I agree to the outlined procedure for plagiarism assessment and potential sanctioning.

Münster, 02 August 2025

Lorenzo Lazarus